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MIL-M-28787/110B
25 May 1994
SUPERSEDING
MIL-M-28787/110A(EC)
6 May 1980

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RECEIVER, INTERFACE, DIGITAL, KEY CODE MDM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic interface receiver module.

1.2 Classification. This specification covers one class of module. Class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Code 603A, Bldg. 2917, NAVSURFWARCENDIV, 300 Highway 361, Crane, IN 47522-5001, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in this issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

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3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 5.28 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be MDM.

3.6.2 Size. The module size shall be 1A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned 1 through 40.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/110-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

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TABLE I Module electrical requirements

Test condition	Parameter	Ref test table	Test type 1/ - 3/	25° ± 5°C				0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	II	X	0 0	5	0 0	5	0 0	5	0.0	5	Ohm
B7,B8	Continuity	II	S	0 0	1	0 0	1	0 0	1	0.0	1	Ohm
C1-C4	Input current, high level	III	X	0 0	6 0	0 0	12	0	9	0	15	µA
C5,C6	Input current, low level	III	X	0 0	4 8	0 0	4 8	0 0	4 8	0 0	4 8	mA
C7-C12	Input current, low level	III	X	0 7	3 0	0 7	3 0	0 7	3 0	0 7	3 0	mA
C13-C18	Input current, high level	III	X	0 0	90	0 0	115	0	700	0	700	µA
D1,D2	Enable voltage, low level	IV	X	0 1	1 0	0 1	1 0	0 1	1 0	0 1	1 0	V
D3-D8	Enable voltage, low level threshold	IV	X	3 8	5 5	3 5	5 5	3 8	5 5	3 5	5 5	V
E1-E12	Output voltage, high level	V	X	3 8	5 5	3 5	5 5	3 8	5 5	3 5	5 5	V
E13-E24	Output voltage, low level	V	X	0 0	0 4	0 0	0 45	0 0	0 42	0 0	0 47	V
F1-F18	Transition time (tTLH)	VI	X	100	350	100	350	100	350	100	350	nS
F19-F30	Transition time (tTHL)	VI	X	6	50	6	50	6	50	6	50	nS
F31-F36	Transition time (tTHL)	VI	X	6	125	6	150	6	150	6	175	nS
G1-G6	Turn-on time (tON)	VII	X	0	500	0	500	0	500	0	500	nS
G7-G12	Turn-on time (tON)	VII	X	210	970	175	1025	150	1050	100	1100	nS
G13-G18	Turn-on time (tON)	VII	X	210	870	175	925	150	950	100	1000	nS
G19-G24	Turn-off time (tOFF)	VII	X	210	870	175	925	150	950	100	1000	nS
G25-G30	Turn-off time (tOFF)	VII	X	0	100	0	100	0	100	0	100	nS
G31-G36	Turn-off time (tOFF)	VII	X	0	500	0	500	0	500	0	500	nS
H1-H6	Output voltage, low level	VIII	X	0 0	0 4	0 0	0 45	0 0	0 42	0 0	0 47	V
H7-H12	Output pulse height	VIII	X	3 5	5 5	3 5	5 5	3 5	5 5	3.5	5 5	V
J1-J6	Transition time (tTHL), low duty cycle	IX	X	6	50	6	50	6	50	6	50	nS
J7-J12	Transition time (tTLH), low duty cycle	IX	X	100	350	100	350	100	350	100	350	nS

-See footnotes at end of table

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TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/ - 3/	25° ± 5°C				0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-Life		
				Min	Max	Min	Max	Min	Max	Min	Max	
J13-J18	Turn-on time (tON), low duty cycle	IX	X	0	500	0	500	0	500	0	500	nS
J19-J24	Turn-off time (tOFF), low duty cycle	IX	X	210	870	175	925	150	950	100	1000	nS
J25-J30	Input current, low duty cycle	IX	X	0	6	0	6	0	6	0	6	mA
K1-K12	Output short circuit current	X	S	0 7	3 0	0 7	3 0	0 7	3 0	0 7	3 0	mA
L1,L2	Power supply current	XI	X	0	70	0	70	0	70	0	70	mA
M	Zener applied potential	XII	S	0	70	0	70	0	70	0	70	mA
N1-N16	Vout (low loaded)	XIII	S	0 0	1 1	0 0	1 1	0 0	1 1	0 0	1 1	V
P1-P4	Input capacitance	XIV	Q	0	50	0	50	0	50	0	50	pF
Q1-Q54	Input transient immunity	XV	S	2.0	5.5	2 0	5 5	2 0	5 5	2 0	5 5	V
R1-R4	Input crosstalk	XVI	Q	0.0	0 6	0 0	0 6	0 0	0.6	0.0	0 6	Vp-p
S1-S16	Output crosstalk	XVII	Q	0 0	0 8	0 0	0 8	0 0	0 8	0 0	0 8	V
T1,T2	Power supply transient rise time	XVIII	Q	20	136	20	136	20	136	20	136	μS
U1-U6	Enable time delay (tdON)	XIX	X	0	870	0	925	0	950	0	1000	nS
U7-U12	Enable time delay (tdOFF)	XIX	X	0	100	0	100	0	100	0	100	nS
V1-V12	Input noise immunity (common mode)	XX	Q	2 0	5 5	2 0	5 5	2 0	5 5	2 0	5 5	V

-See footnotes on next page

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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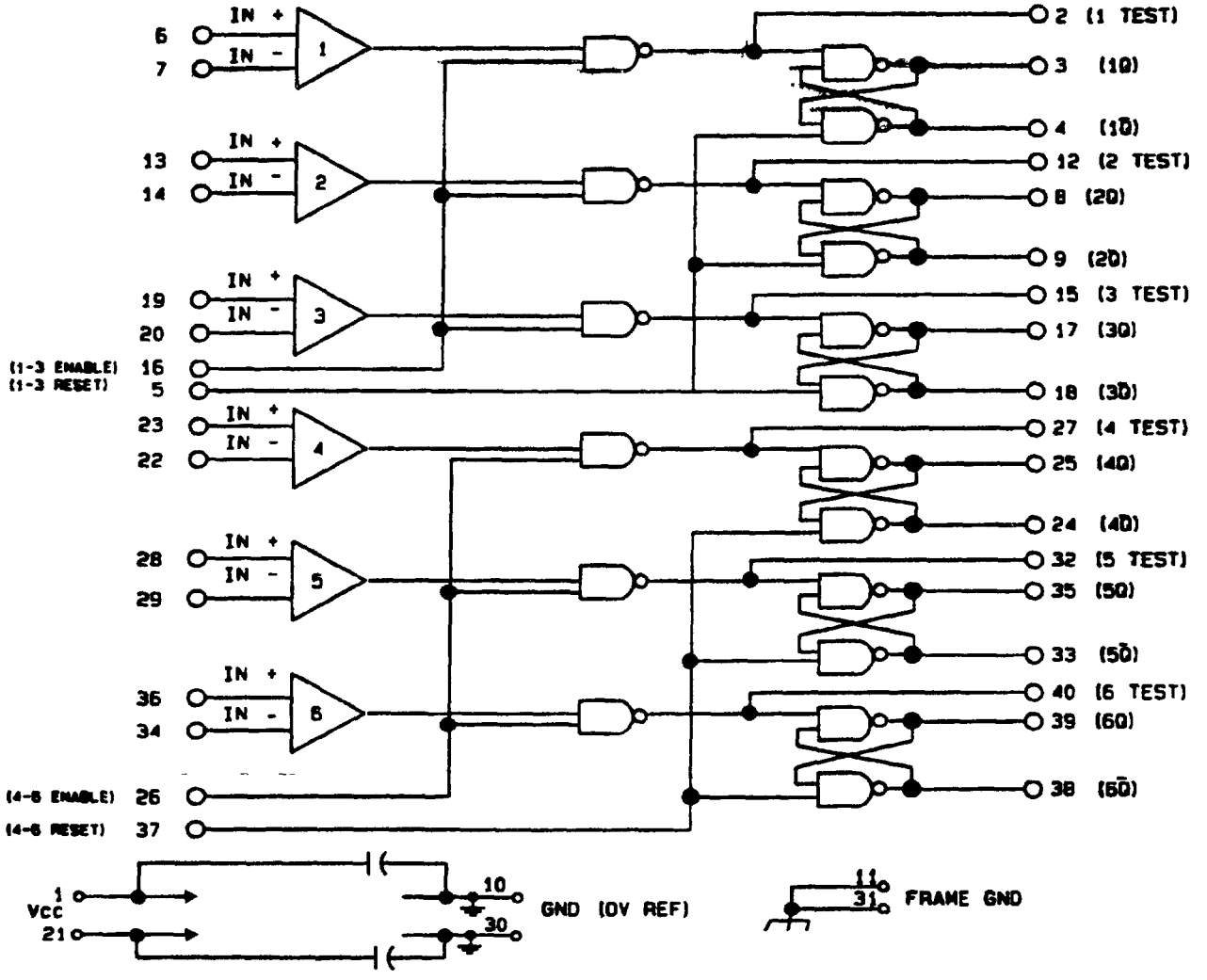


FIGURE 1. MDM function diagram, interface receiver.

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3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (VCC) of 5.0 ± 0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ± 1 percent.

3.11 Input noise immunity. The receiver circuits shall have 0.5 volt/microseconds of differential input noise immunity and +200 volts and -200 volts of common mode input noise immunity with less than ±50 percent overshoot and undershoot, and a rise time of 10 nanoseconds or less.

3.12 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. Test conditions A, B, K, M, and N are stress tests and shall be performed only at 25° ± 5°C. If test conditions A, B, K, M, or N are part of a sequence of tests, the order of testing shall be as

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follows: A, B, K, M, and N where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

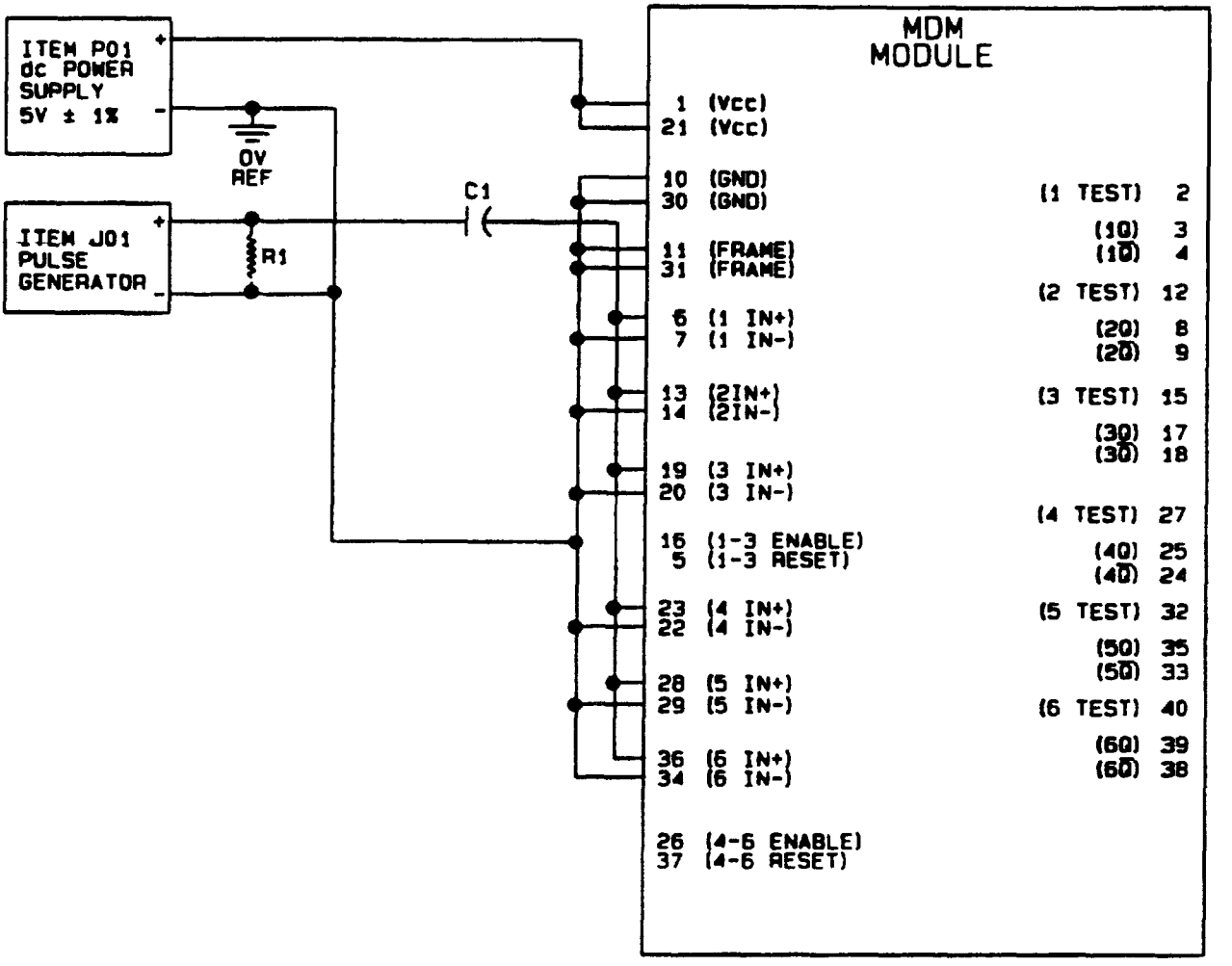
4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through J, L, Q, and U shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 2. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ± 5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through J, L, Q, and U shall be performed in the order specified in 4.4.2 using the 0 to 60°C operating temperature end-of-life requirements in table I. Following the return of the module to 25° ± 5°C, electrical tests C through J, L, Q, and U shall be performed in the order specified in 4.4.2 using the 25° ± 5°C ambient end-of-life requirements in table I.

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NOTES:

1. R1 = 50 Ohms ± 5 percent, 1 W.
2. C1 = 10 µF ± 20 percent, 15 V.
3. Item J01 setup:
   PW = 1.5 ± 0.25 µS.
   PA = 5 ± 0.5 V.
   PRP = 15 ± 5 µS.
   tTLH = tTHL = 50 ± 25 nS.
4. Resistor R1 and capacitor C1 are located at the module end of the generator signal cable.

FIGURE 2. Operating temperature, component temperature, vibration and life test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Cross reference of classification. Military specification MIL-M-28787/110 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658833. Part number M28787/110-1 supersedes part numbers 2658833 and M28787/110.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits in table I.

Supply voltage	5 Vdc ± 10 percent
Maximum power dissipation	385 mW
Input voltage	
Low level	0.0 V minimum, 1.2 V maximum
High level	
DATA	2.4 V minimum, 5.5 V maximum
ENABLE, RESET, and TEST	1.8 V minimum, 5.5 V maximum

Input current	
Low level (VIL = 0.0V)	
ENABLE and RESET	4.8 mA max
TEST	0.7 mA min. 3.0 mA max
High level	
DATA (VIH = 5.5V)	6.0 mA max
ENABLE and RESET (VIH = 4.0V)	15 µA max
TEST (VIH = 5.5V)	0 µA min. 700 µA max

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Output voltage	
Low level	0.0V min, 0.47V max
High level	3.5 min, 5.5V max
Output current	
Q and $\overline{Q}$	
Low level	9.6 mA max
High level	55 μA max
TEST	
Low level	9.6 mA max
High level	55 μA max
Turn-on time	
DATA to Q	0 nS min, 500 nS max
DATA to $\overline{Q}$	100 nS min, 1100 nS max
DATA to TEST	100 nS min, 1000 nS max
Turn-off time	
DATA to Q	100 nS min, 1000 nS max
DATA to $\overline{Q}$	0 nS min, 100 nS max
DATA to TEST	0 nS min, 500 nS max
Enable time delay	
tdON	0 nS min, 1000 nS max
tdOFF	0 nS min, 100 nS max
Data input pulse width	1.1 μs min, 1.5 μs max
Common mode rejection	± 100 V

6.5 Module function. This module contains six receiver circuits with transformer coupled inputs; the output of each circuit is configured as a flip-flop. There are two ENABLE inputs and two RESET inputs. Each ENABLE and RESET input services three receiver circuits. Each receiver circuit has a TEST terminal which may be used in a maintenance loop to test the corresponding output flip-flop. If the ENABLE terminal is at a low level, the input of the receiver will be inhibited and the TEST terminal will remain at a high level. The flip-flop circuit can be set by applying a low level at the TEST terminal. The ENABLE terminal must be at a low level in order to set the flip-flop with the TEST terminal. The output of the flip-flop can be reset by applying a low level to the RESET terminal. The receiver circuits will respond to a 2.4 to 5.5 volt, 1.1 to 1.5 μS differential input signal at pulse repetition rates up to 167 kHz. This module generates standard 5.0 volt logic levels which appear as a negative going pulse at the TEST terminal which sets the output of a flip-flop.

6.5.1 Input to output relationship. Each interface receiver has the input to output relationship for positive logic as shown on figure 3.

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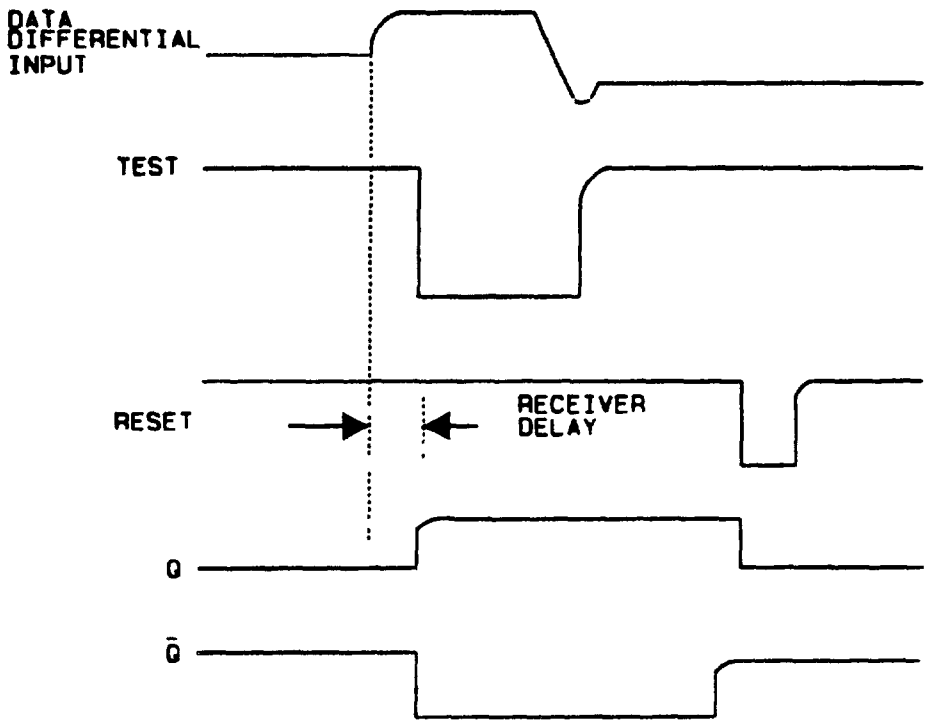


FIGURE 3. Waveform of input to output relationship.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Army - ER
Navy - NW
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

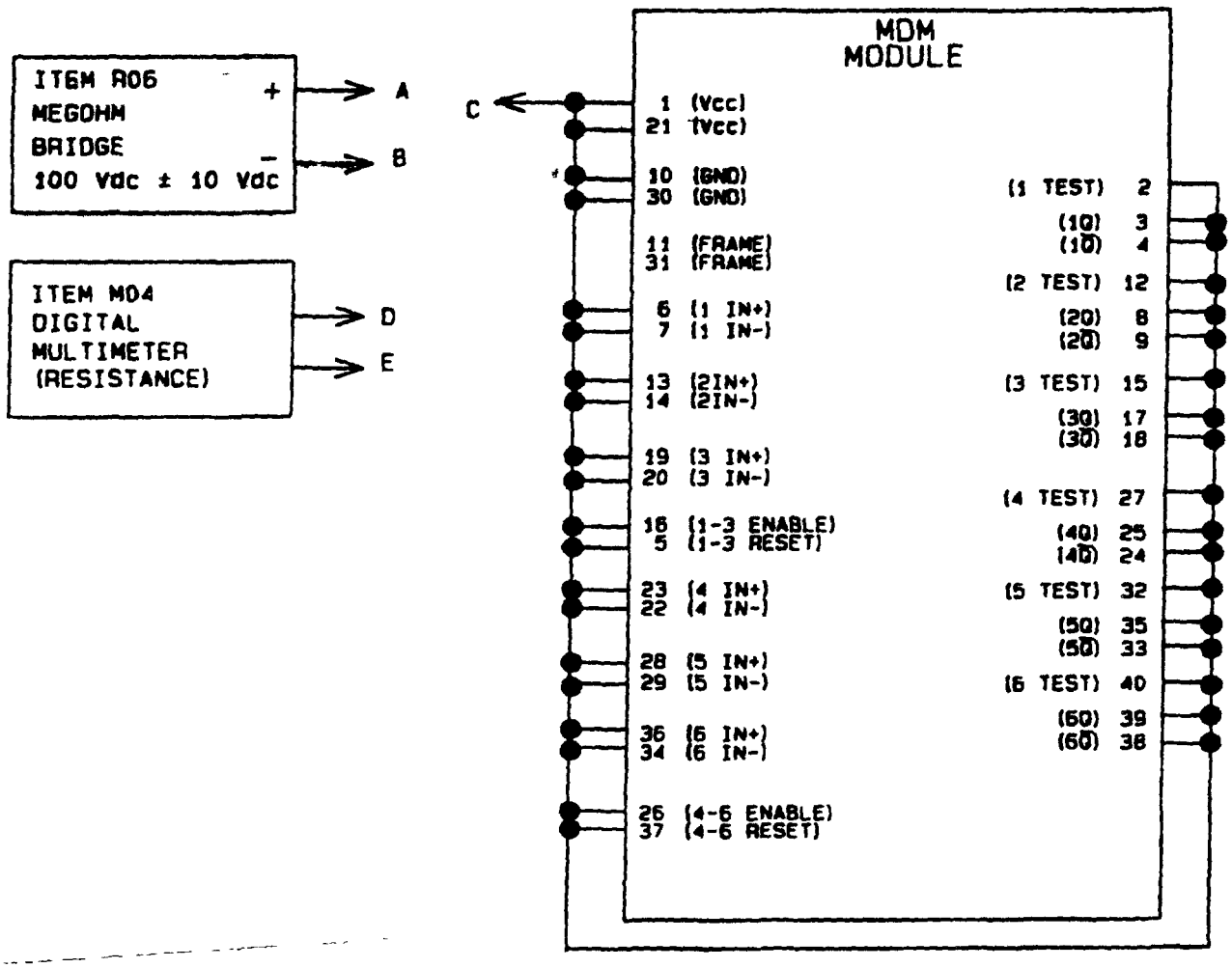
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TABLE II. Isolation and continuity test.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to	B to	D to	E to	Min	Max		
A	Isolation	C	11,31	Open	Open	10	INF	MOhm	4
B1	Continuity	Open	Open	6	7	0.0	5.0	Ohm	
B2				13	14				
B3				19	20				
B4				23	22				
B5				28	29				
B6				36	34		5.0		
B7				31	Frame		1.0		
B8	Continuity	Open	Open	11	Frame	0.0	1.0	Ohm	4

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

NOTE: For test conditions B7 and B8, penetrate the anodized finish on the module with a sharply pointed probe connected to the negative terminal.

FIGURE 4. Isolation and continuity test circuit for table II.

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TABLE III. Input current (high-level and low-level) test.

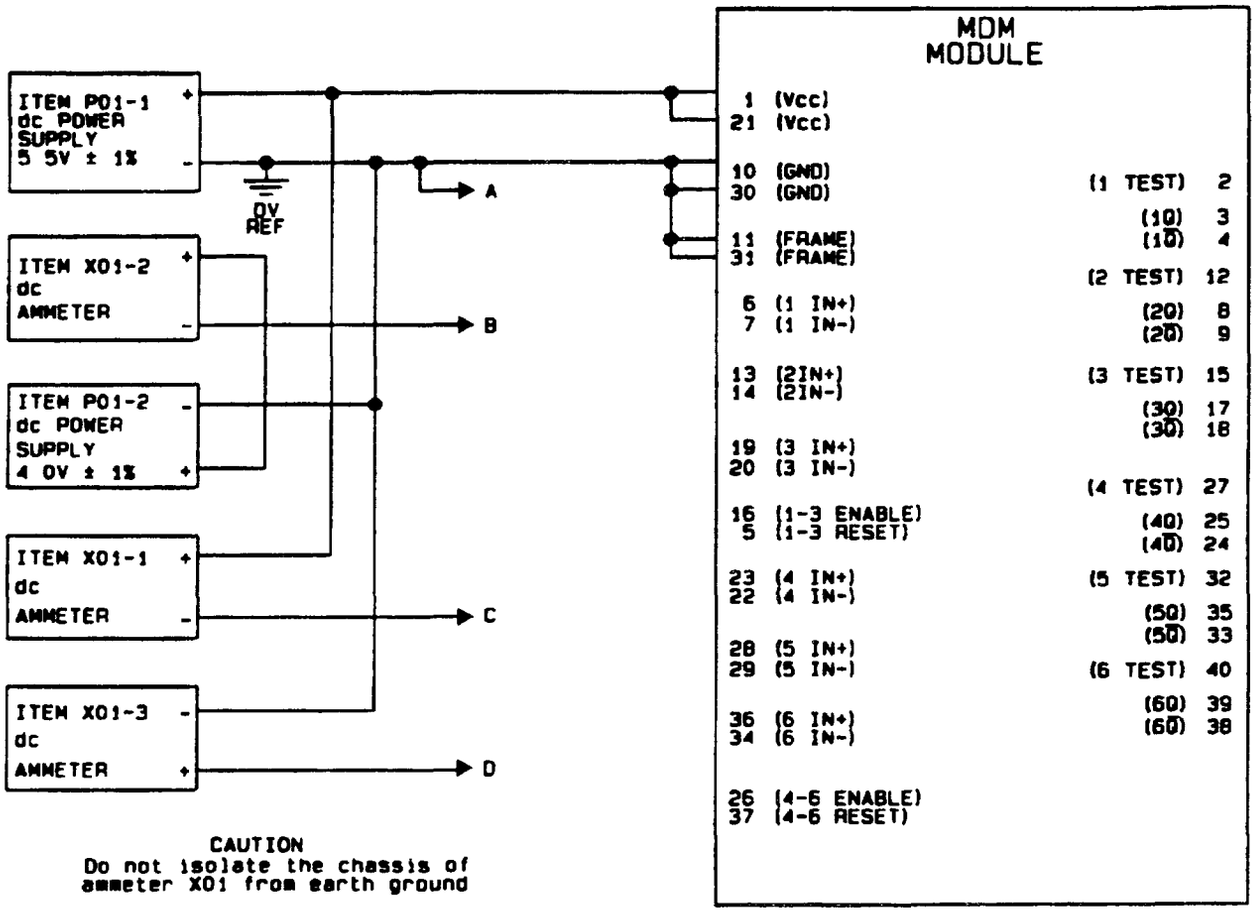
Test cond	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	Min	Max		
C1	Input current, high level	Open	16	Open	Open	0.0	6.0	µA	5
C2		Open	26						
C3		3,8,17	5						
C4	Input current, high level	25,35, 39	37		Open		6.0	µA	
C5	Input current, low level	2,12 15	Open		5		4.8	mA	
C6		27,32, 40			37	0.0	4.8		
C7		5			2	0.7	3.0		
C8					12				
C9		5			15				
C10		37			27				
C11					32				
C12	Input current, low level	37		Open	40	0.7	3.0	mA	
C13	Input current, high level	4		2	Open	0.0	90	µA	
C14	Input current, high level	9	Open	12	Open	0.0	90	µA	5

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TABLE III. Input current (high-level and low-level) test - Continued.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	Min	Max		
C15	Input current, high level	18	Open	15	Open	0.0	90	µA	5
C16		24		27					
C17		33		32					
C18	Input current, high level	38	Open	40	Open	0.0	90	µA	5

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NOTE: Only one item X01 is required since items X01-1, X01-2, and X01-3 are not used simultaneously.

FIGURE 5. Input current (high-level and low-level) test circuit for table III.

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TABLE IV. Enable voltage (low-level and low-level threshold) test.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Min	Max		
D1	Enable voltage, low level	16	Open	Open	0.1	1.0	V	6,7
D2	Enable voltage, low level	26	Open	Open	0.1	1.0		
D3	Enable voltage, low level threshold	Open	2	16	3.8	5.5		
D4			12					
D5			15	16				
D6			27	26				
D7			32					
D8	Enable voltage, low level threshold	Open	40	26	3.8	5.5	V	6,7

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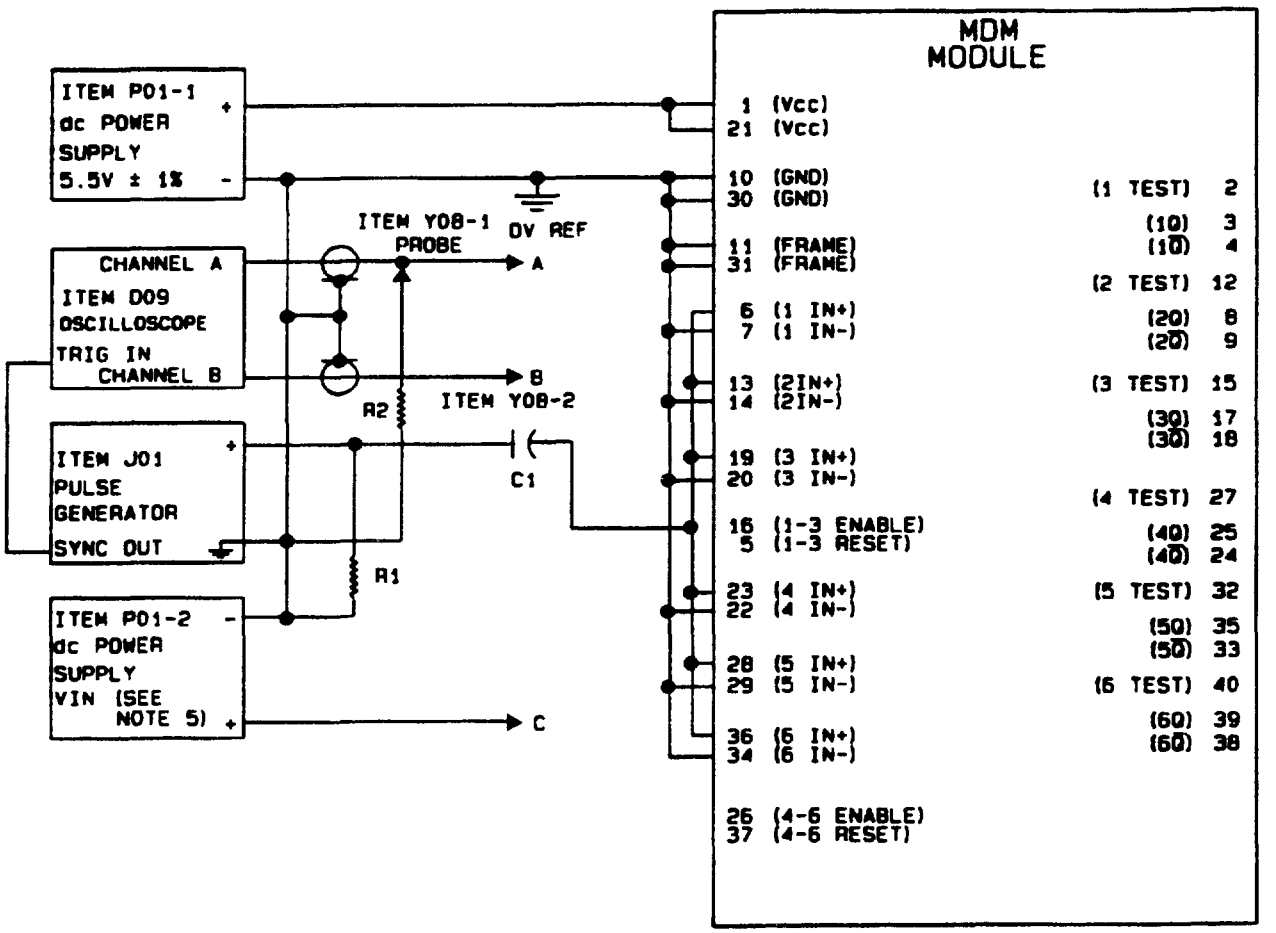


FIGURE 6. Enable voltage (low-level and low-level threshold) test circuit for table IV.

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NOTES:

1. R1 = 50 Ohms ± 5 percent, 1 W.

2. R2 = 200 Ohms ± 1 percent, 1/4 W.

3. C1 = 10 µF ± 20 percent, 15 Vdc.

4. Resistor R1 and capacitor C1 are located at the module end of the generator signal cable.

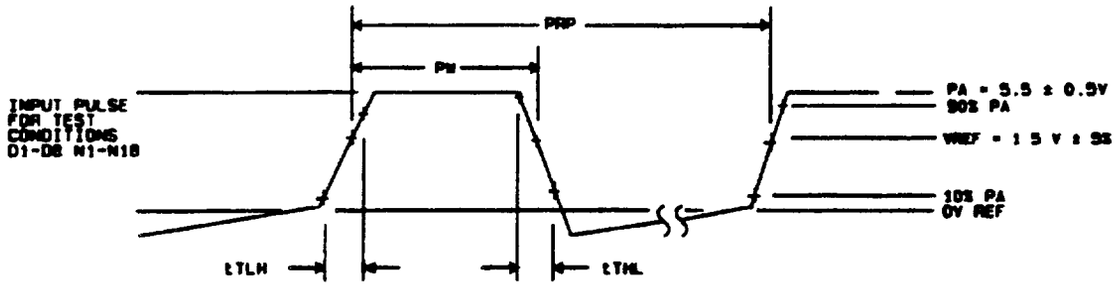
5. Use appropriate VIN for specified test conditions and temperature listed below:

Test condition	Temperature (°C)	VIN(V)
D3-D8	0	1.20
	25	1.10
	60	.95
	80	.80

0°, 60°, and 80°C refer to operating temperatures (module fin); 25°C refers to ambient temperature. For 80°C, use the 0 to 60°C end-of-life limits.

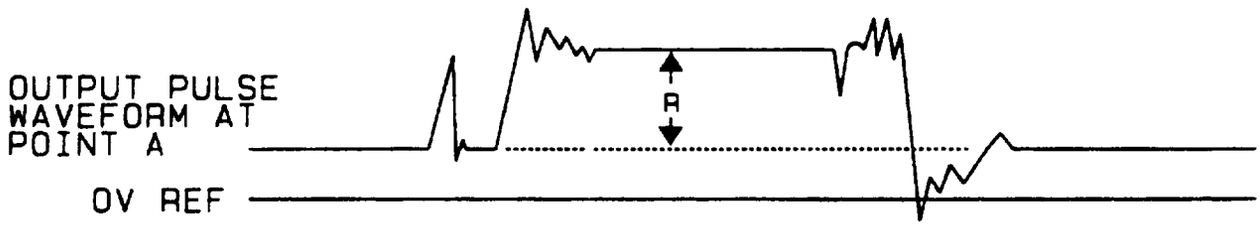
FIGURE 6. Enable voltage (low-level and low-level threshold) test circuit for table IV - Continued.

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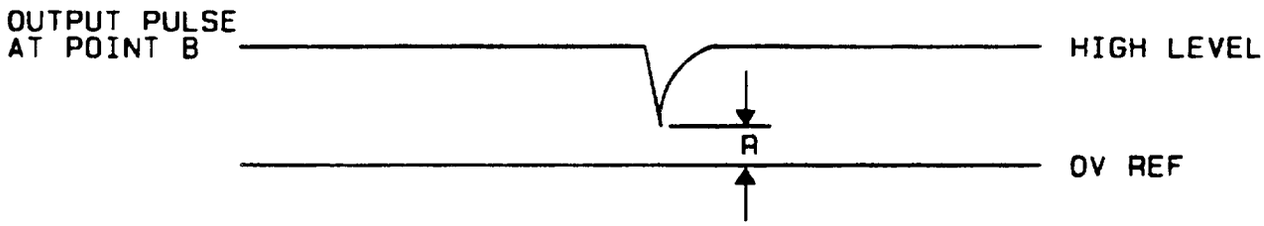
NOTES:

1. Slight pulse degradation may occur due to generator loading (droop, backswing).
2. Input pulse parameters:
   PW = 1.5 ± 0.2 µS.
   PRP = 7.0 ± 0.5 µS.
   tTLH = tTHL = 40 ± 10 nS.



NOTE:

1. The value R shall conform to the requirements of test conditions D1 and D2.



NOTE:

1. The measured value of R shall conform to the requirements of test conditions D3 through D8.

FIGURE 7. Enable voltage (low-level and low-level threshold) and VOUT (low-loaded) test circuit waveforms for tables IV and XIII.

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TABLE V. Output voltage (high-level and low-level) test.

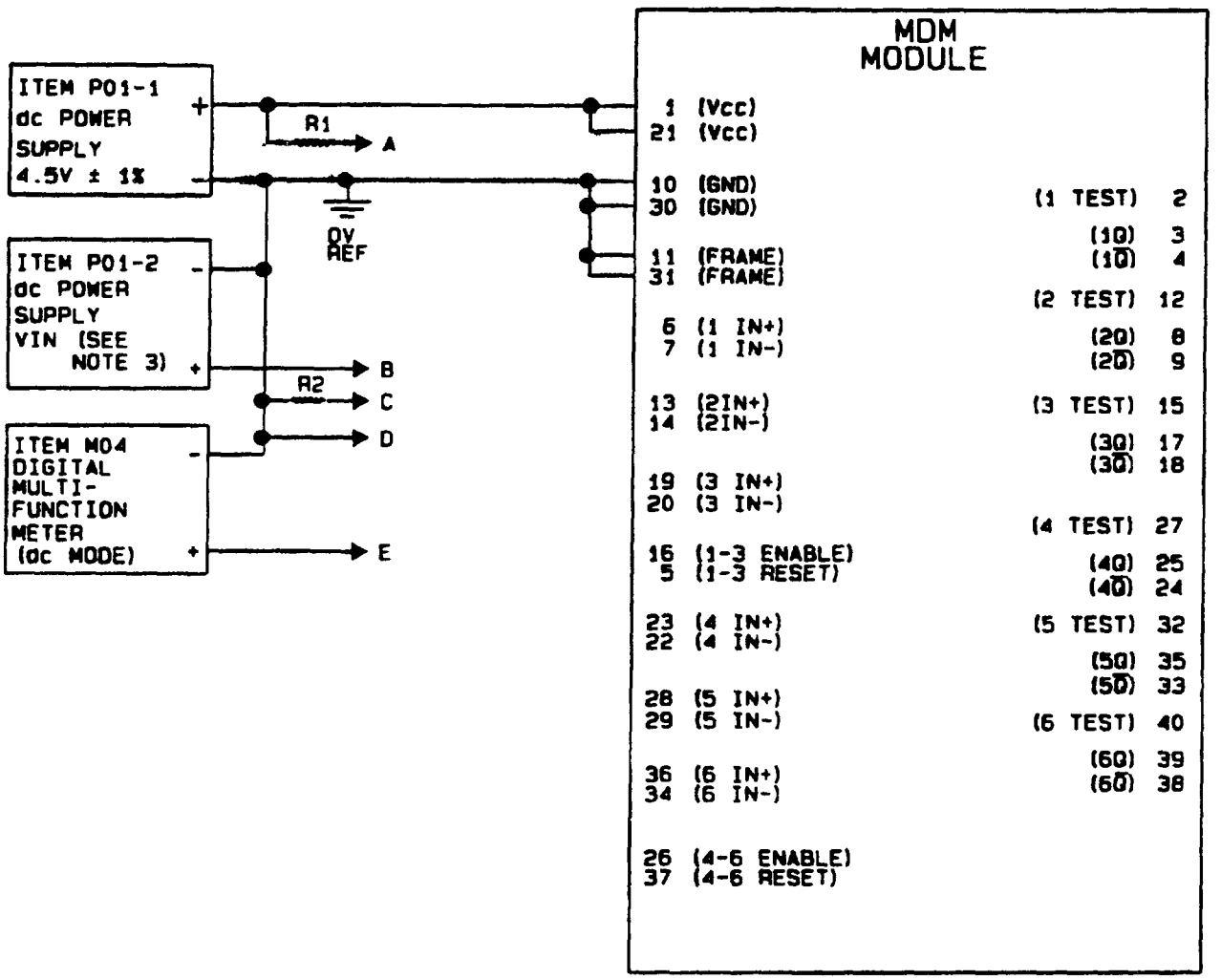
Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25° ± 5°C Ambient Min	Initial limits 25° ± 5°C Ambient Max	Unit	Test fig.
E1	$1\overline{Q}$ Output voltage, high level	Open	5	4	2	4	3.8	5.5	V	8
E2	$2\overline{Q}$			9	12	9				
E3	$3\overline{Q}$		5	18	15	18				
E4	$4\overline{Q}$		37	24	27	24				
E5	$5\overline{Q}$			33	32	33				
E6	$6\overline{Q}$		37	38	40	38				
E7	1Q		2	3	5	3				
E8	2Q		12	8		8				
E9	3Q		15	17	5	17				
E10	4Q		27	25	37	25				
E11	5Q		32	35	37	35				
E12	6Q Output voltage, high level	Open	40	39	37	39	3.8	5.5		
E13	1Q Output voltage, low level	3	2	Open	5	3	0.0	0.4		
E14	2Q	8	12			8				
E15	3Q	17	15		5	17				
E16	4Q Output voltage, low level	25	27	Open	37	25	0.0	0.4	V	8

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TABLE V. Output voltage (high-level and low-level) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25° ± 5°C Ambient Min	Initial limits 25° ± 5°C Ambient Max	Unit	Test fig.
E17	5Q Output voltage, low level	35	32	Open	37	35	0.0	0.4	V	8
E18	6Q	39	40		37	39				
E19	$1\overline{Q}$	4	5		2	4				
E20	$2\overline{Q}$	9			12	9				
E21	$3\overline{Q}$	18	5		15	18				
E22	$4\overline{Q}$	24	37		27	24				
E23	$5\overline{Q}$	33			32	33				
E24	$6\overline{Q}$ Output voltage, low level	38	37	Open	40	38	0.0	0.4	V	8

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NOTES:

1. R1 = 348 Ohms ± 1 percent, 1/4 W.
2. R2 = 61.9 KOhms ± 1 percent, 1/8 W.

FIGURE 8. Output voltage (high-level and low-level) test circuit for table V.

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NOTES: - Continued.

3. Use appropriate VIN for specified test condition and temperature listed below:

Test condition	Temperature (°C)	VIN
E1-E12	0	1.2
	25	1.1
	60	.95
	80	.80
E13-E24	0	2.0
	25	1.9
	60	1.8
	80	1.8

0°, 60°, and 80°C refer to operating temperatures (module fin); 25°C refers to ambient temperature. For 80°C, use the 0 to 60°C end-of-life limits.

FIGURE 8. Output voltage (high-level and low-level) test circuit for table V - Continued.

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TABLE VI. Transition time (tTLH and tTHL) test.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	1Q Transition time (tTLH)	6	16	26	3	100	350	nS	9,10
F2	2Q	13			8				
F3	3Q	19	16	26	17				
F4	4Q	23	26	16	25				
F5	5Q	28			35				
F6	6Q	36	26	16	39				9,10
F7	$1\overline{Q}$	6	16	26	4				9,11
F8	$2\overline{Q}$	13			9				
F9	$3\overline{Q}$	19	16	26	18				
F10	$4\overline{Q}$	23	26	16	24				
F11	$5\overline{Q}$	28			33				
F12	$6\overline{Q}$ Transition time (tTLH)	36	26	16	38				9,11
F13	1 Test Transition time (tTLH)	6	16	26	2				9,12
F14	2	13			12				
F15	3	19	16	26	15				
F16	4	23	26	16	27				
F17	5 Test Transition time (tTLH)	28	26	16	32	100	350	nS	9,12

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TABLE VI. Transition time (tTLH and tTHL) test - Continued.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F18	6 Test Transition time (tTLH)	36	26	16	40	100	350	nS	9,12
F19	1Q Transition time (tTHL)	6	16	26	3	6	50		9,10
F20	2Q	13			8				
F21	3Q	19	16	26	17				
F22	4Q	23	26	16	25				
F23	5Q	28			35				
F24	6Q	36	26	16	39				9,10
F25	$1\overline{Q}$	6	16	26	4				9,11
F26	$2\overline{Q}$	13			9				
F27	$3\overline{Q}$	19	16	26	18				
F28	$4\overline{Q}$	23	26	16	24				
F29	$5\overline{Q}$	28			33				
F30	$6\overline{Q}$ Transition time (tTHL)	36	26	16	38		50		9,11
F31	1 Test Transition time (tTHL)	6	16	26	2		125		9,12
F32	2 Test Transition time (tTHL)	13	16	26	12	6	125	nS	9,12

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TABLE VI. Transition time (tTLH and tTHL) test - Continued.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F33	3 Test Transition time (tTHL)	19	16	26	15	6	125	nS	9,12
F34	4	23	26	16	27				
F35	5	28			32				
F36	6 Test Transition time (tTHL)	36	26	16	40	6	125	nS	9,12

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TABLE VII. Turn-on (tON) and turn-off (tOFF) time test.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1Q Turn-on time (tON)	6	16	26	3	0	500	nS	9,10
G2	2Q	13			8				
G3	3Q	19	16	26	17				
G4	4Q	23	26	16	25				
G5	5Q	28			35				
G6	6Q	36	26	16	39	0	500		9,10
G7	$1\overline{Q}$	6	16	26	4	210	970		9,11
G8	$2\overline{Q}$	13			9				
G9	$3\overline{Q}$	19	16	26	18				
G10	$4\overline{Q}$	23	26	16	24				
G11	$5\overline{Q}$	28			33				
G12	$6\overline{Q}$ Turn-on time (tON)	36	26	16	38		970		9,11
G13	1 Test Turn-on time (tON)	6	16	26	2		870		9,12
G14	2	13			12				
G15	3	19	16	26	15				
G16	4	23	26	16	27				
G17	5	28			32				
G18	6 Test Turn-on time (tON)	36	26	16	40	210	870	nS	9,12

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TABLE VII. Turn-on (tON) and turn-off (tOFF) time test - Continued.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G19	1Q Turn-off time (tOFF)	6	16	26	3	210	870	nS	9,10
G20	2Q	13			8				
G21	3Q	19	16	26	17				
G22	4Q	23	26	16	25				
G23	5Q	28			35				
G24	6Q Turn-off time (tOFF)	36	26	16	39	210	870		9,10
G25	1$\overline{Q}$ Turn-off time (tOFF)	6	16	26	4	0	100		9,11
G26	2$\overline{Q}$	13			9				
G27	3$\overline{Q}$	19	16	26	18				
G28	4$\overline{Q}$	23	26	16	24				
G29	5$\overline{Q}$	28			33				
G30	6$\overline{Q}$ Turn-off time (tOFF)	36	26	16	38		100		9,11
G31	1 Test Turn-off time (tOFF)	6	16	26	2		500		9,12
G32	2	13			12				
G33	3	19	16	26	15				
G34	4 Test Turn-off time (tOFF)	23	26	16	27	0	500	nS	9,12

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TABLE VII. Turn-on (tON) and turn-off (tOFF) time test - Continued.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G35	5 Test Turn-off time (tOFF)	28	26	16	32	0	500	nS	9,12
G36	6 Test Turn-off time (tOFF)	36	26	16	40	0	500	nS	9,12

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TABLE VIII. Output voltage (low-level) and output pulse height test.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	1 Test Output voltage, low level	6	16	26	2	0.0	0.4	V	9,12
H2	2	13			12				
H3	3	19	16	26	15				
H4	4	23	26	16	27				
H5	5	28			32				
H6	6 Test Output voltage, low level	36	26	16	40	0.0	0.4		
H7	1 Test Output pulse height	6	16	26	2	3.5	5.5		
H8	2	13			12				
H9	3	19	16	26	15				
H10	4	23	26	16	27				
H11	5	28			32				
H12	6 Test Output pulse height	36	26	16	40	3.5	5.5	V	9,12

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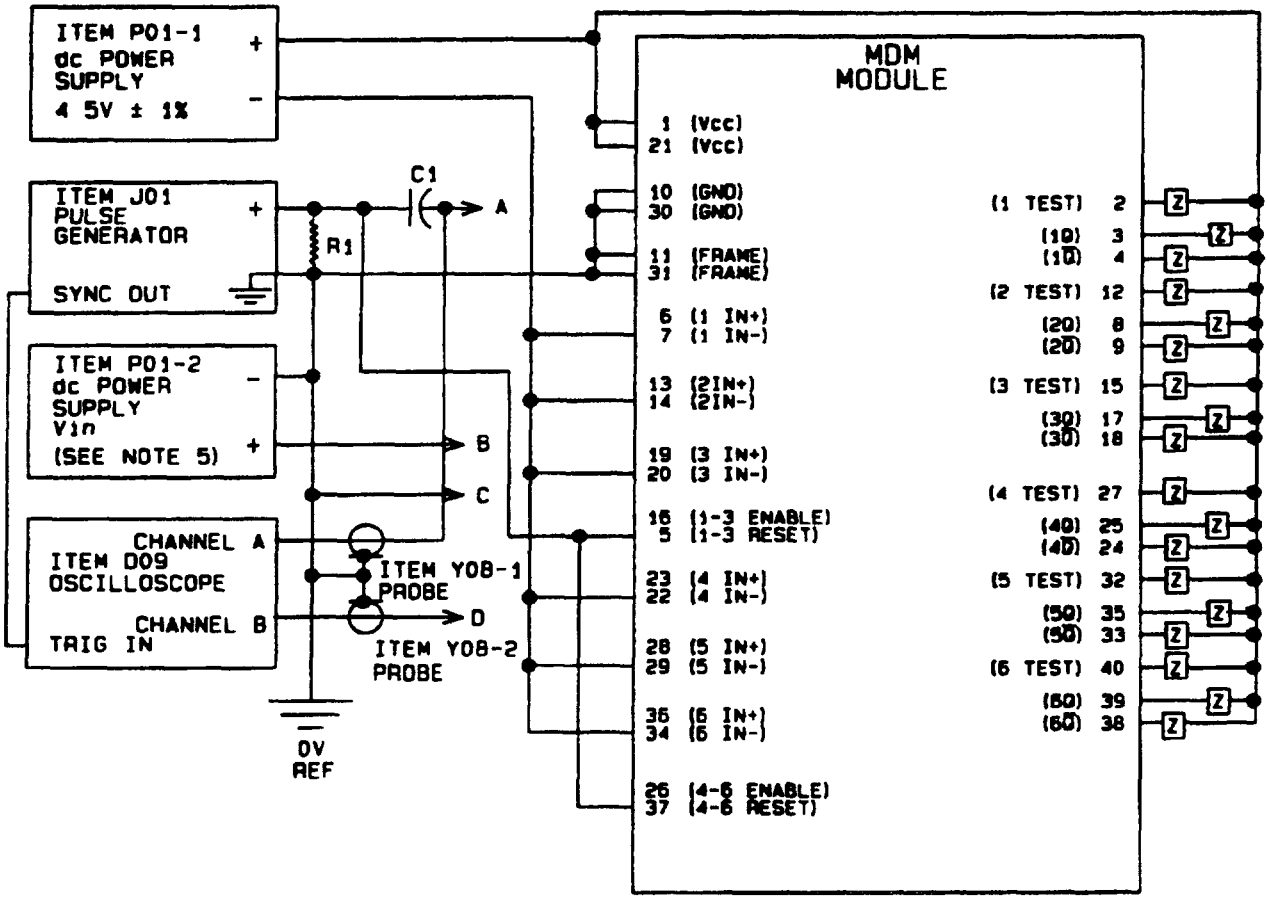


FIGURE 9. Transition time (tTLH and tTHL), turn-on and turn-off time (tON and tOFF), output voltage (low-level) and output pulse height test circuit for tables VI, VII, and VIII.

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NOTES:

1. R1 = 50 Ohms ± 5 percent, 1 W.

2. C1 = 10 μF ± 20 percent, 15 Vdc.

3. Load all outputs per figure 13.

4. Resistor R1 and capacitor C1 are located at the module end of the generator signal cable.

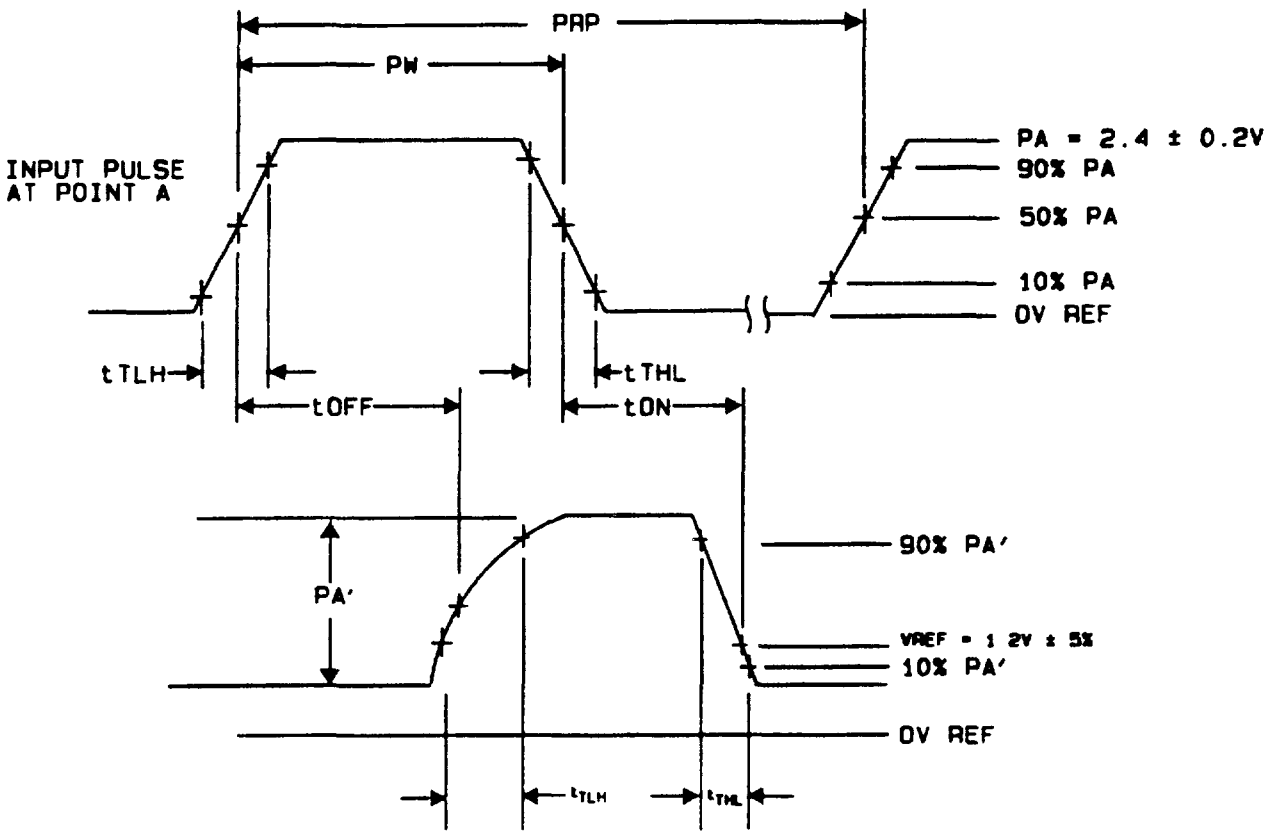
5. Use appropriate VIN for specified test condition and temperature listed below:

Test condition	Temperature (°C)	VIN (volts)
F1-F36, G1-G36, H1-H12	0	2.0
	25	1.9
	60	1.8
	80	1.8

0°, 60°, and 80°C refer to operating temperatures (module fin); 25°C refers to ambient temperature. For 80°C, use the 0 to 60°C end-of-life limits.

FIGURE 9. Transition time (tTLH and tTHL), turn-on and turn-off time (tON and tOFF), output voltage (low-level) and output pulse height test circuit for tables VI, VII, and VIII - Continued.

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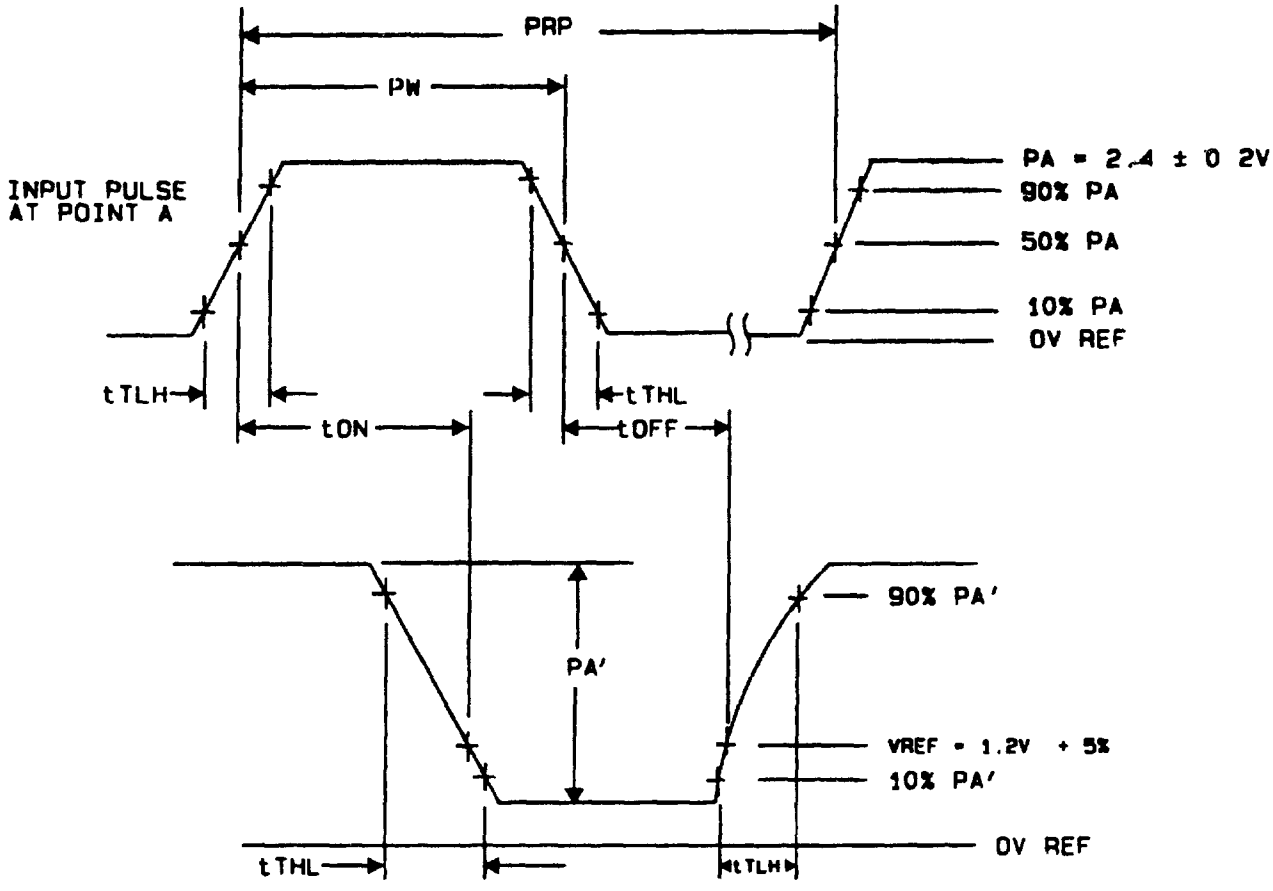
NOTE:

1. Input pulse parameters:

   PW = 1.15 ± 0.05 µS.
   PRP = 6.0 ± 0.4 µS.
   tTLH = tTHL = 20 ± 4 nS.

FIGURE 10. Transition time (tTLH and tTHL), turn-on and turn-off time (tON and tOFF), and power supply current test circuit waveforms for tables VI, VII, and XI.

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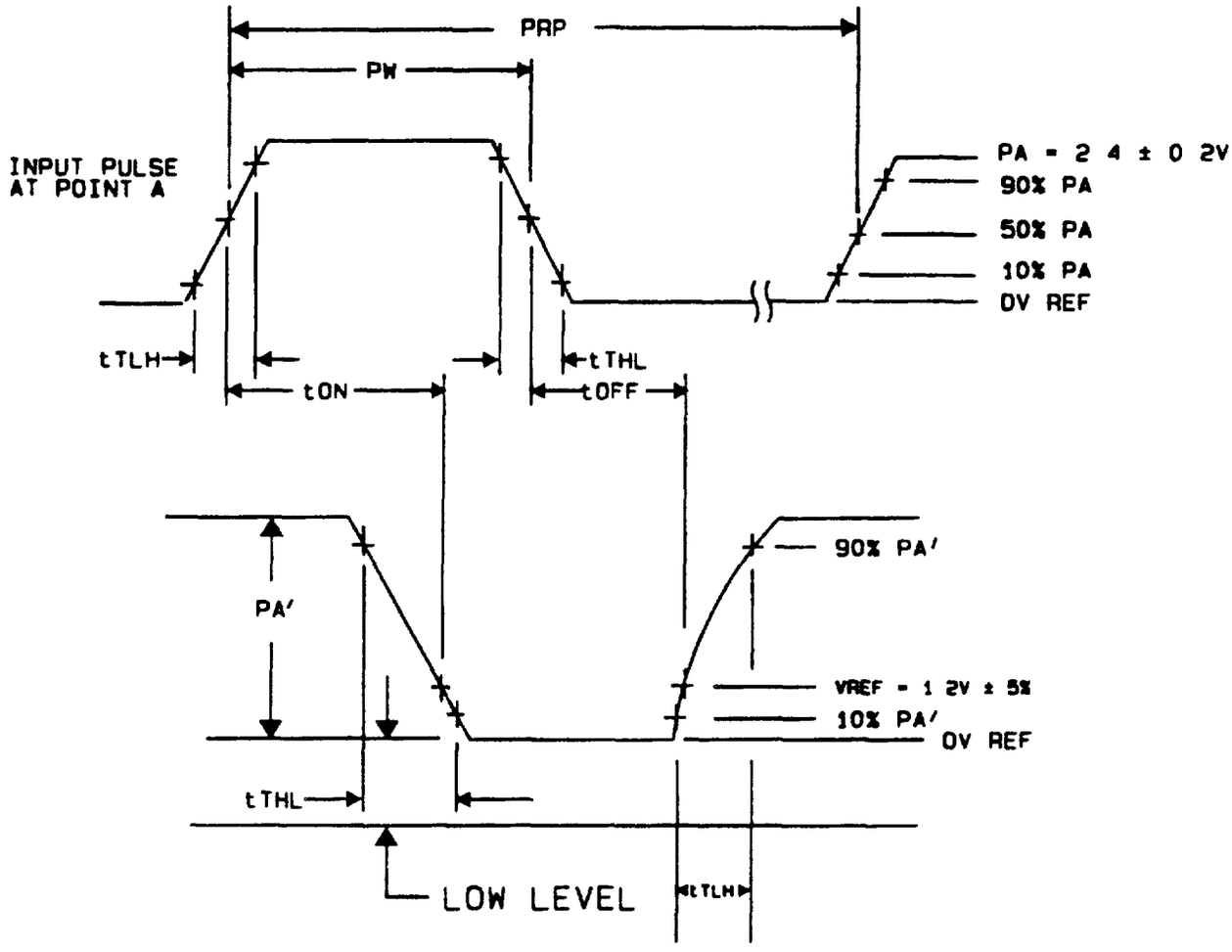
NOTE:

1. Input pulse parameters:

   PW = 1.15 ± 0.05 µS.
   PRP = 6 ± 0.4 µS.
   tTLH = tTHL = 20 ± 4 nS.

FIGURE 11. Transition time (tTLH and tTHL) and turn-on and turn-off time (tON and tOFF) test circuit waveforms for tables VI and VII.

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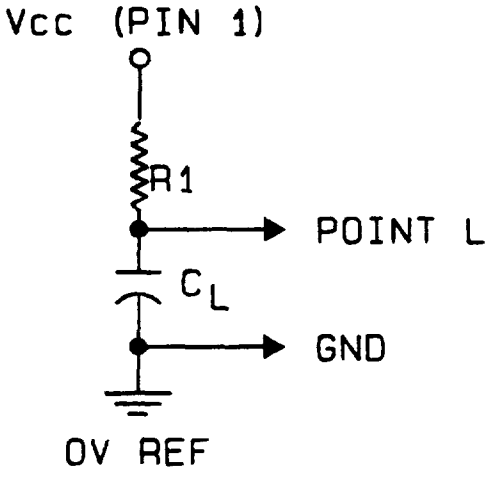
NOTE:

1. Input pulse parameters:

   PW = 1.15 ± 0.05 µS.
   PRP = 6 ± 0.4 µS.
   tTLH = tTHL = 20 ± 4 nS.

FIGURE 12. Transition time (tTLH and tTHL), turn-on and turn-off time (tON and tOFF), output voltage (low-level), output pulse height and VOUT (low-loaded) test circuit waveforms for tables VI, VII, VIII, and XIII.

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NOTES:

1. R1 = 3.83 KOhms ± 1 percent, 1/8 W for test conditions F, G, H, J, Q, U, and V.
   R1 = 422 Ohms ± 1 percent, 1/8 W for test condition S.

2. CL = 30 ± 5 pF (including test connector, probe and compensating capacitance if required. If the addition of a compensating capacitor is required, it shall be connected directly at the test connector pins).

3. With the module removed from the test connector, measure the capacitance between the test connector pins listed in the following table. The measurements shall be performed at the point of mechanical contact between the module blade pin and the test connector pin. The measured values at 1 MHz ± 30 percent shall be as specified therein. For these measurements, disconnect the R1 resistors from the respective output pins. The resistors should be disconnected as close to their bodies as practical. Reconnect these resistors prior to using the test connector for module testing.

FIGURE 13. MDM logic load (Z) circuit.

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NOTES: - Continued.

Test connector pins		
From pin	To pins	CL (see note 2)
2	10,11,30,31,7,14,20,22,29,34	30 ± 5 pF
12		
15		
27		
32		
40		
3		
4		
8		
9		
17		
18		
25		
24		
35		
33		
39		
38	10,11,30,31,7,14,20,22,29,34	30 ± 5 pF

4. The loop length from pin 1 to each output, including the length of the body of the resistor, shall not exceed 4.5 inches (114.3 mm).

FIGURE 13. MDM logic load (Z) circuit - Continued.

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TABLE IX. Transition time (tTHL and tTLH), turn-on and turn-off time (tON and tOFF) and input current, low duty cycle test.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Min	Max		
J1	Q1 Transition time (tTHL), low duty cycle	6	6	3	6	50	nS	14,15
J2	Q2	13	13	8				
J3	Q3	19	19	17				
J4	Q4	23	23	25				
J5	Q5	28	28	35				
J6	Q6 Transition time (tTHL), low duty cycle	36	36	39	6	50		
J7	Q1 Transition time (tTLH), low duty cycle	6	6	3	100	350		
J8	Q2	13	13	8				
J9	Q3	19	19	17				
J10	Q4	23	23	25				
J11	Q5	28	28	35				
J12	Q6 Transition time (tTLH), low duty cycle	36	36	39	100	350		
J13	Q1 Turn-on time (tON), low duty cycle	6	6	3	0	500		
J14	Q2 Turn-on time (tON), low duty cycle	13	13	8	0	500	nS	14,15

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TABLE IX. Transition time (tTHL and tTLH), turn-on and turn-off time (tON and tOFF) and input current, low duty cycle test - Continued.

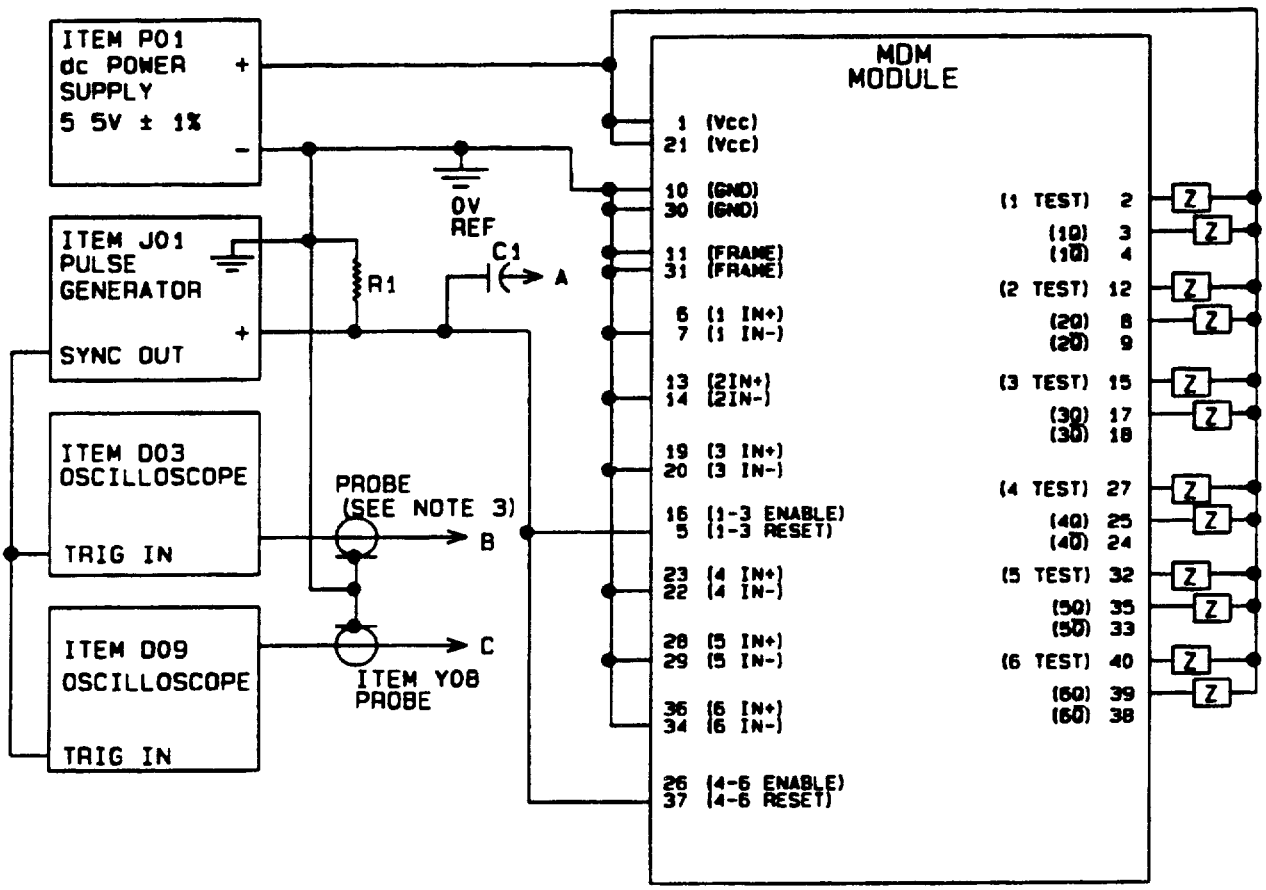
Test cond	Parameter	Connect			Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Min	Max		
J15	Q3 Turn-on time (tON), low duty cycle	19	19	17	0	500	nS	14,15
J16	Q4	23	23	25				
J17	Q5	28	28	35				
J18	Q6 Turn-on time (tON), low duty cycle	36	36	39	0	500		
J19	Q1 Turn-off time (tOFF), low duty cycle	6	6	3	210	870		
J20	Q2	13	13	8				
J21	Q3	19	19	17				
J22	Q4	23	23	25				
J23	Q5	28	28	35				
J24	Q6 Turn-off time (tOFF), low duty cycle	36	36	39	210	870	nS	
J25	Q1 Input current, low duty cycle	6	6	3	0	6	mA	
J26	Q2	13	13	8				
J27	Q3	19	19	17				
J28	Q4 Input current, low duty cycle	23	23	25	0	6	mA	14,15

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TABLE IX. Transition time (tTHL and tTLH), turn-on and turn-off time (tON and tOFF) and input current, low duty cycle test - Continued.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Min	Max		
J29	Q5 Input current, low duty cycle	28	28	35	0.0	6	mA	14,15
J30	Q6 Input current, low duty cycle	36	36	39	0.0	6	mA	14,15

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NOTES:

1. C1 = 10 μF ± 20 percent, 5 Vdc.

2. R1 = 50 Ohms ± 5 percent, 1 W.

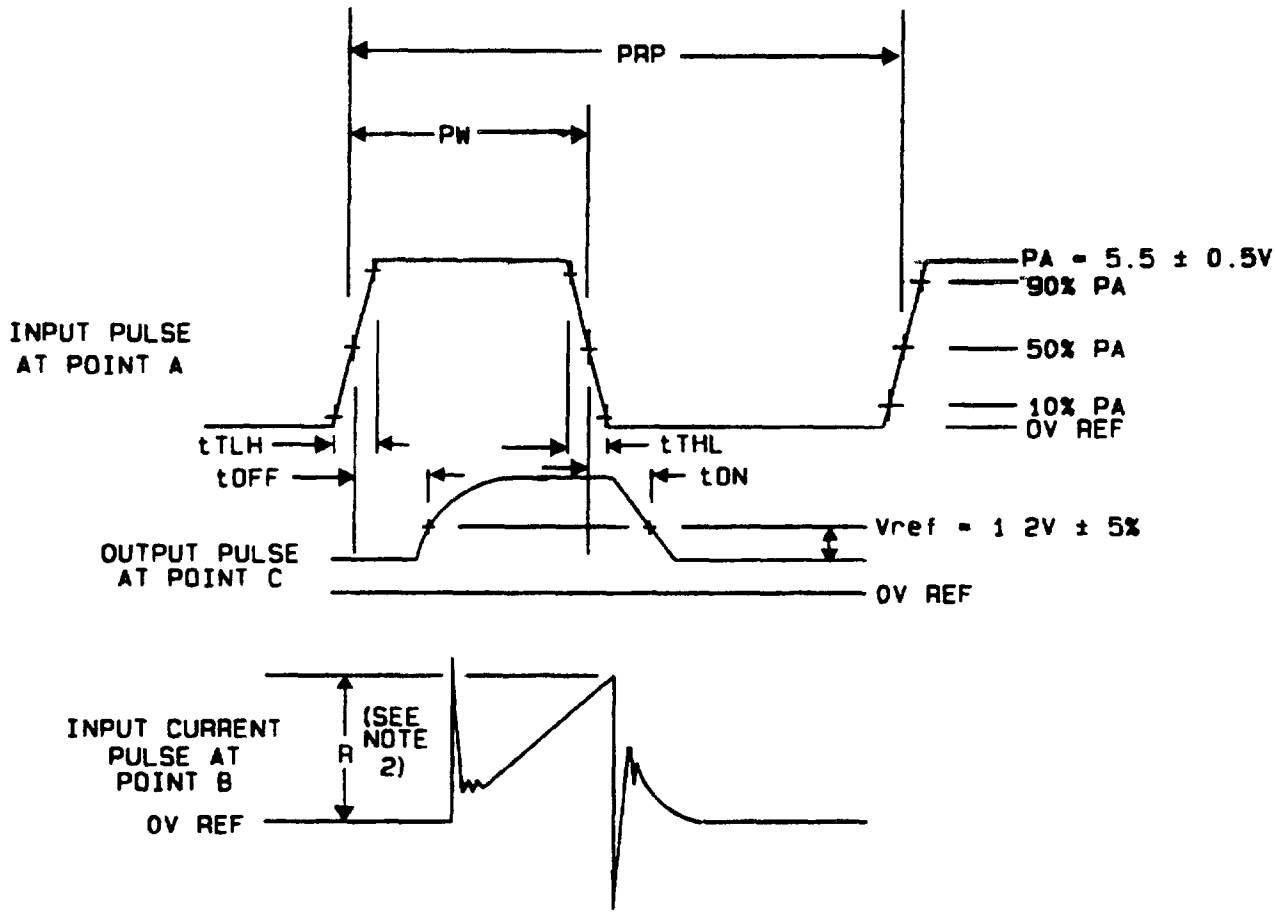
3. For test conditions J1-J24, use item Y08 and for test conditions J25-J30, use item Y03 placed over wire making the connection between the module pin and capacitor C1.

4. Load outputs per figure 13.

5. Resistor R1 and capacitor C1 are located at the module end of the generator signal cable.

FIGURE 14. Transition time (tTHL and tTLH), turn-on and turn-off (tON and tOFF) and input current, low duty cycle test circuit for table IX.

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NOTES:

1. Input pulse parameters:

   tTLH = tTHL = 20 ± 4 nS.
   PRP = 100 ± 10 µS.
   PW = 1.45 ± 0.05 µS for test conditions J1-J24.
   PW = 2.5 µS ± 10 percent for test conditions J25-J30.

2. Turn-on and turn-off current transients may exceed table IX values. The measured value of R shall conform to the requirements of test conditions J25-J30.

FIGURE 15. Transition time (tTHL and tTLH), turn-on and turn-off (tON and tOFF) and input current, low duty cycle test circuit waveforms for table IX.

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TABLE X. Output short circuit current test.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Min	Max		
K1	Output short circuit current	3	2	0.7	3.0	mA	16
K2		4	5				
K3		8	12				
K4		9	5				
K5		17	15				
K6		18	5				
K7		25	27				
K8		24	37				
K9		35	32				
K10		33	37				
K11		39	40				
K12	Output short circuit current	38	37	0.7	3.0	mA	16

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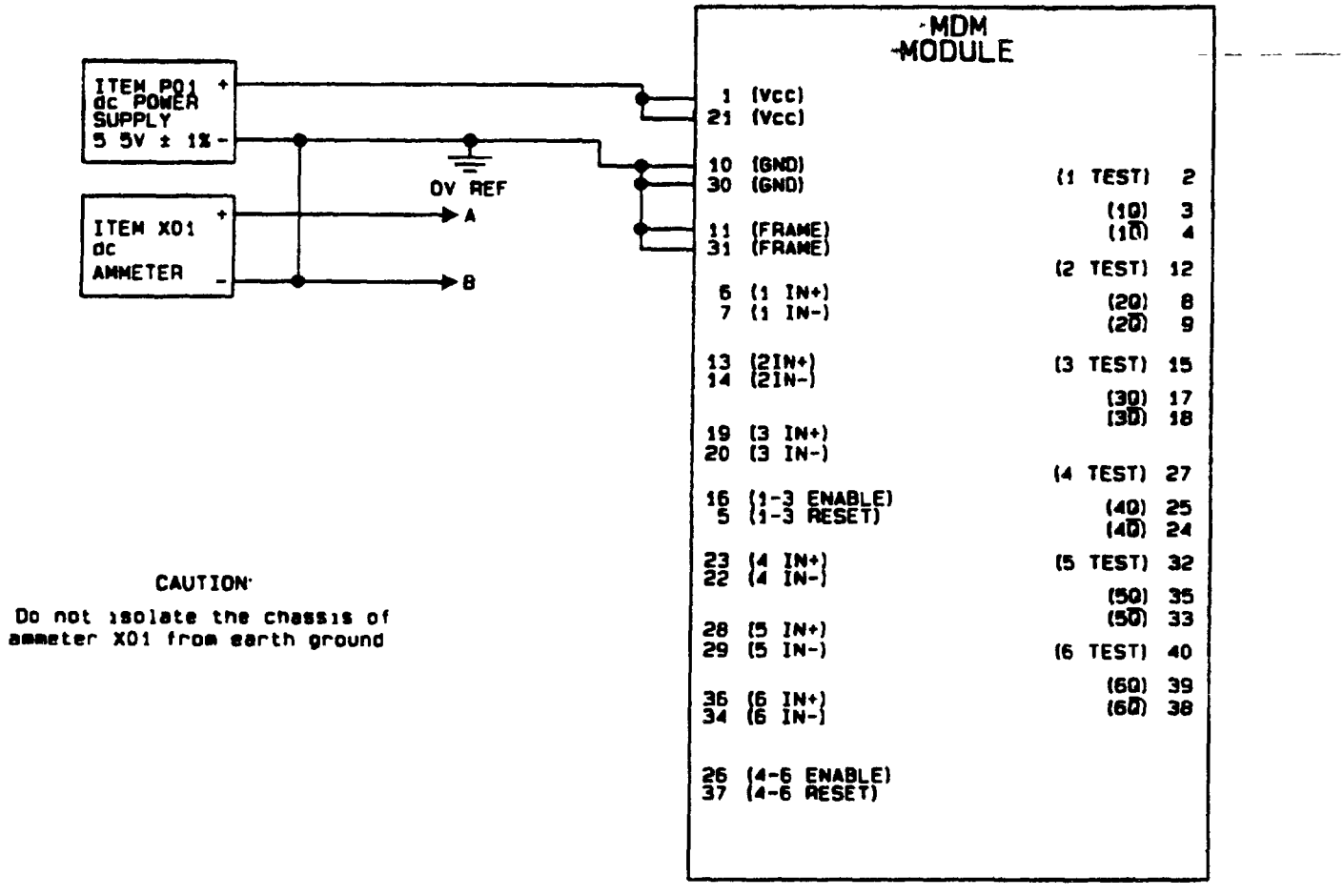


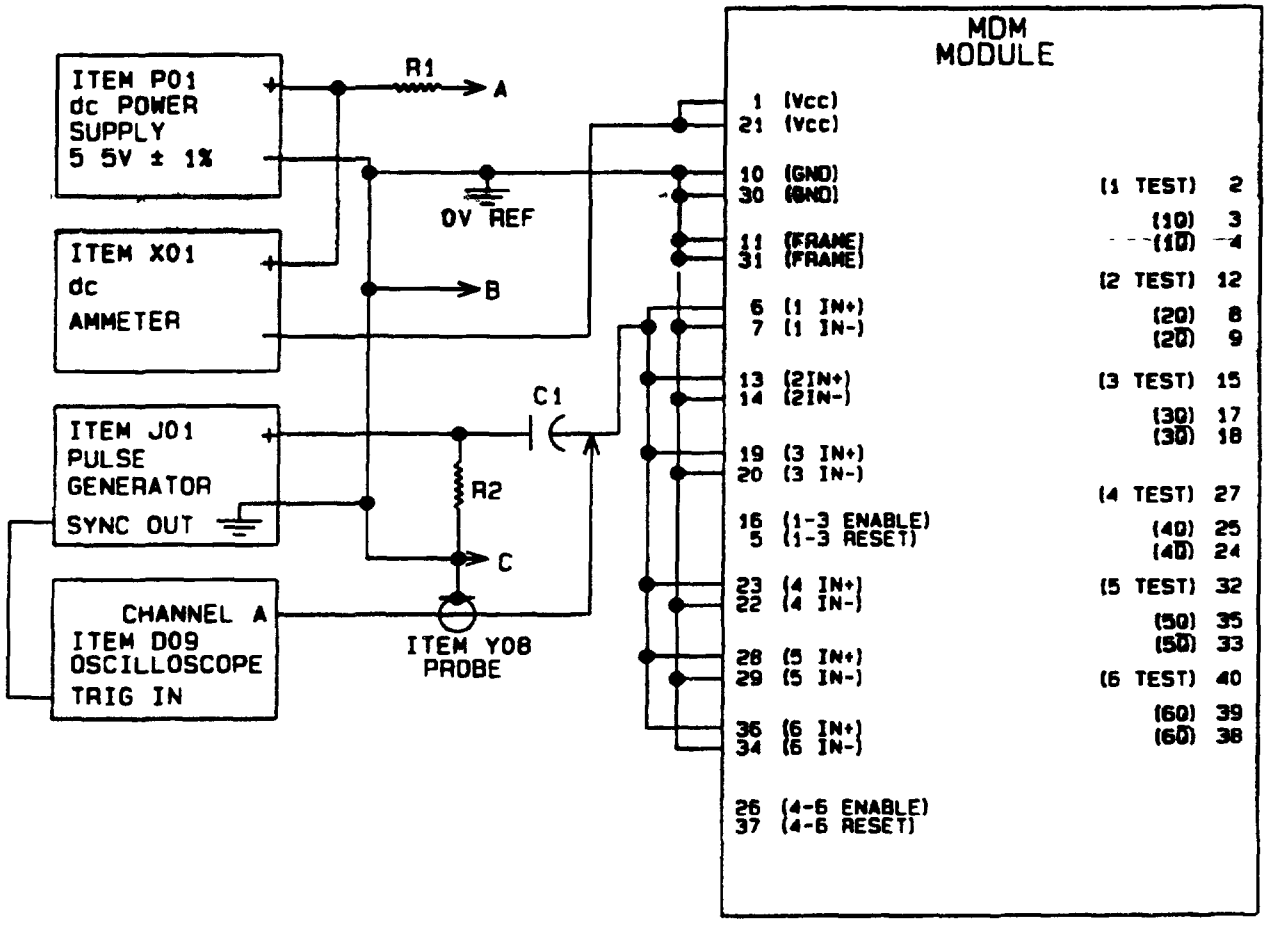
FIGURE 16. Output short circuit current test circuit for table X.

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TABLE XI. Power supply current test.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Min	Max		
L1	Power supply current	16,26	Open	5,37	0	70	mA	10, 17
L2	Power supply current	Open	5,16, 26,37	Open	0	70	mA	10, 17

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NOTES:

1. R1 = 3.1 KOhms ± 1 percent, 1/4 W.

2. R2 = 50 Ohms ± 5 percent, 1 W.

3. C1 = 10 μF ± 20 percent, 15 Vdc.

4. Item J01 shall be adjusted to obtain the waveform shown on figure 10. Slight pulse degradation may occur due to generator loading (droop, backswing).

5. Resistor R2 and capacitor C1 are located at the module end of the generator signal cable.

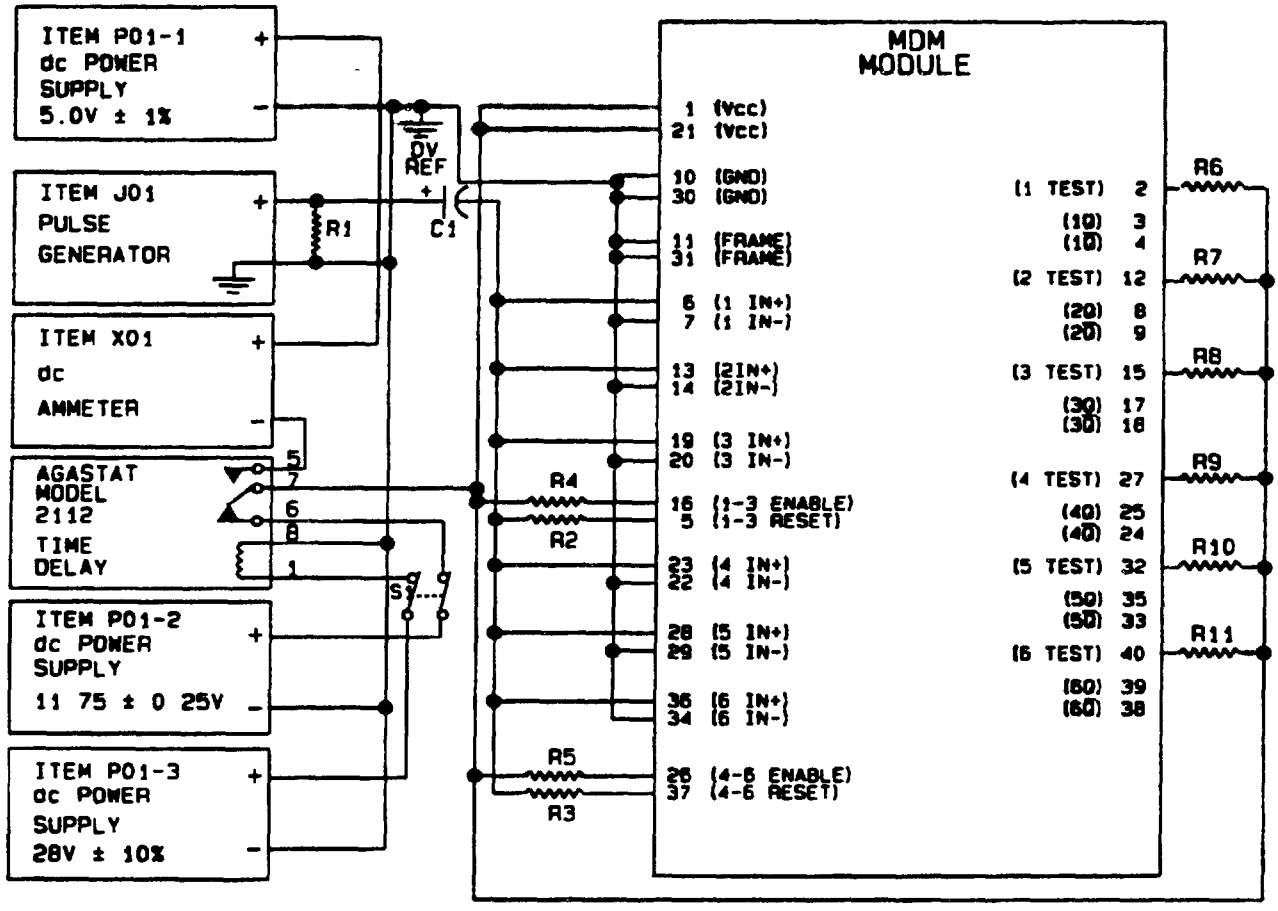
FIGURE 17. Power supply current test circuit for table XI.

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TABLE XII. Zener applied potential test.

Test cond	Parameter	Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		Min	Max		
M	Zener applied potential	0	70	mA	18

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground. If test condition M is successively repeated, the time interval between execution of each test should be a minimum of 10 seconds in order to preclude detrimental effects due to module heating.

FIGURE 18. Zener applied potential test circuit for table XII.

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NOTES:

1. R1 = 50 Ohms ± 5 percent, 1 W.

2. R2, R3 = 750 Ohms ± 1 percent, 1/4 W.

3. R4-R11 = 1 KOhm ± 1 percent, 1/8 W.

4. C1 = 10 μF ± 20 percent, 15 Vdc.

5. S1 = DPST switch.

6. Adjust the time delay relay for a 1 ± 0.2 μS delay. (Agastat Model 2112 relay).

7. Close switch S1 and after 1.2 seconds read ammeter X01. Open switch S1 at the conclusion of the test.

8. Item J01 setup:

   PW = 1.5 ± 0.5 μS.
   PA = 8.5 ± 0.5 V.
   PRP = 15 ± 5 μS.
   tTLH = tTHL = 50 ± 10 nS.

9. Resistor R1 and capacitor C1 are located at the module end of the generator signal cable.

10. Output of item J01 is adjusted before connecting to appropriate module pins. Some pulse degradation may occur when the pulse generator is connected to the module pins.

FIGURE 18. Zener applied potential test circuit for table XII - Continued.

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TABLE XIII. VOUT (low-loaded) test.

Test cond	Parameter	Connect					Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N1	VOUT (low loaded)	5	3	Open	Open	2	0.0	1.1	V	7,19
N2		2	4			5				
N3		5	8			12				
N4		12	9			5				
N5		5	17			15				
N6		15	18			5				
N7		37	25			27				
N8		27	24			37				
N9		37	35			32				
N10		32	33			37				
N11		37	39			40				
N12		40	38	Open	Open	37				7,19
N13		Open	2	6	2	16				7, 12, 19
N14			12	13	12					
N15			15	19	15	16				
N16			27	23	27	26				
N17			32	28	32					
N18	VOUT (low loaded)	Open	40	36	40	26	0.0	1.1	V	7, 12, 19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
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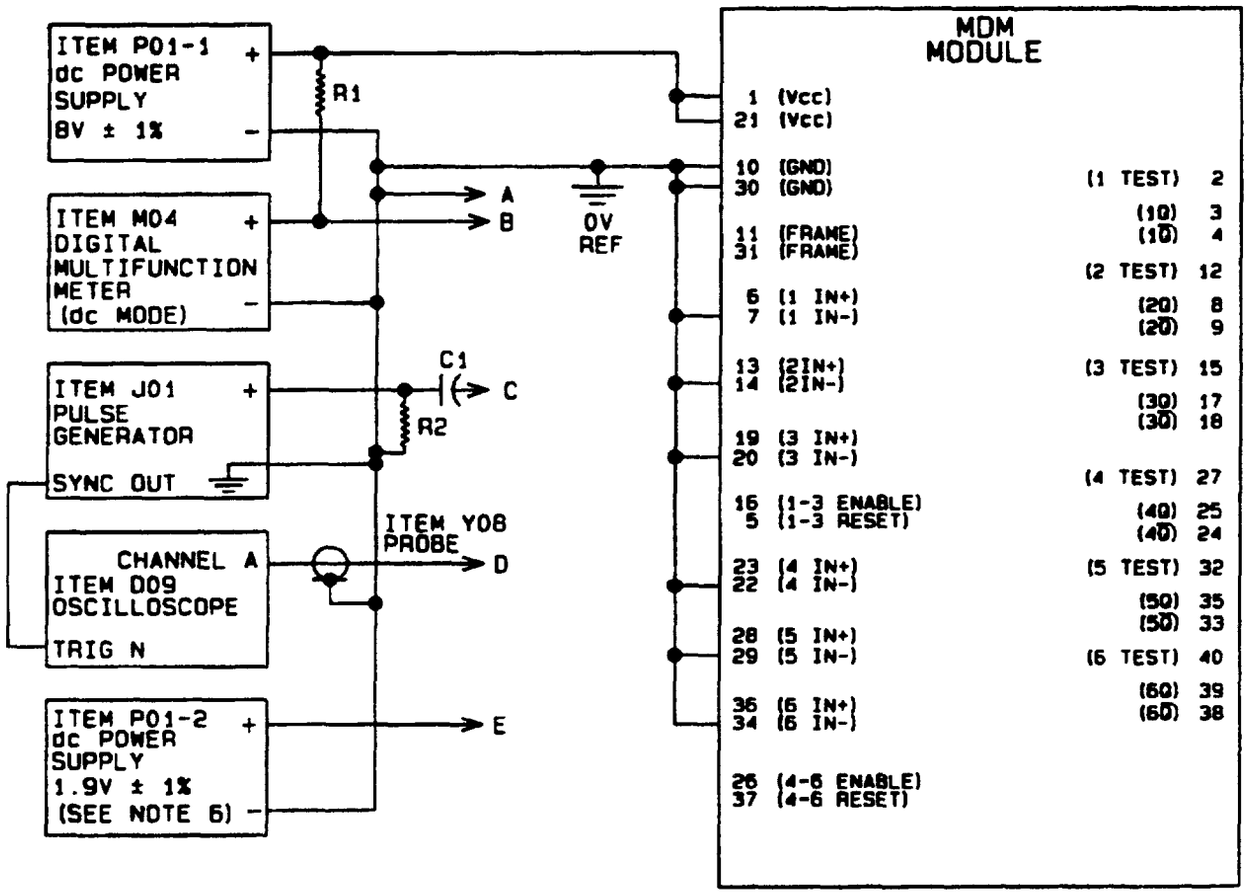


FIGURE 19. VOUT (low-loaded) test circuit for table XIII.

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NOTES:

1. R1 = 274 Ohms ± 1 percent, 1/4 W.

2. C1 = 10 µF ± 20 percent, 15 Vdc.

3. R2 = 50 Ohms ± 5 percent, 1 W.

4. Connect points A and D to module pins before connecting points B and C.

5. The output voltage shall be monitored for a minimum of two minutes after point B or point C have been connected to the module pins.

6. The voltage given for item P01-2 is for 25°C; for 0°C operating temperature use 2.0 volts; for 60°C operating temperature, use 1.8 volts.

7. Resistor R2 and capacitor C1 are located at the module end of the generator signal cable.

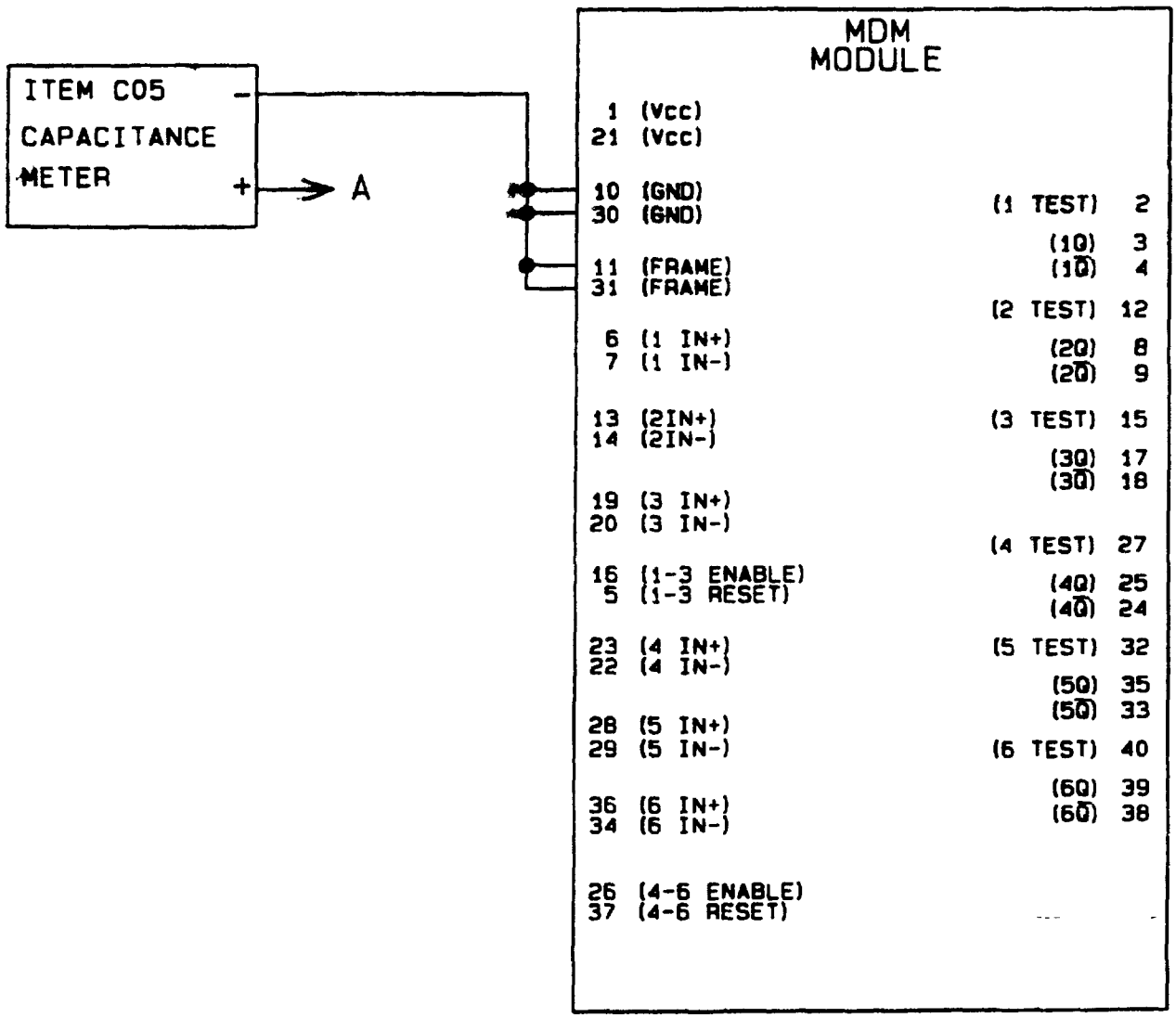
FIGURE 19. VOUT (low-loaded) test circuit for table XIII - Continued.

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TABLE XIV. Input capacitance test.

Test cond	Parameter	Connect A to pin	Initial limits 25° ± 5°C Ambient Min	Initial limits 25° ± 5°C Ambient Max	Unit	Test fig.
P1	Input capacitance	16	0	50	pF	20
P2		5				
P3		26				
P4	Input capacitance	37	0	50	pF	20

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NOTES:

1. Do not attach test connector or wires to any of the module pins other than pins 10, 11, 30, and 31 and the pin under test. Stray capacitance added to other pins may affect the readings.

2. Null out the effects of the test leads prior to connecting to the module.

FIGURE 20. Input capacitance test circuit for table XIV.

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TABLE XV. Input transient immunity test.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	Input transient immunity	Open	2	3	5	2.0	5.5	V	21, 22
Q2			5	4	2				
Q3			12	8	5				
Q4			5	9	12				
Q5			15	17	5				
Q6			5	18	15				
Q7			27	25	37				
Q8			37	24	27				
Q9			32	35	37				
Q10			37	33	32				
Q11			40	39	37				
Q12		Open	37	38	40				
Q13		6	16	2	Open				
Q14		13		12					
Q15		19	16	15					
Q16		23	26	27					
Q17		28		32					
Q18		36	26	40					
Q19	Input transient immunity	6	Open	2	Open	2.0	5.5	V	21, 22

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TABLE XV. Input transient immunity test - Continued.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q20	Input transient immunity	6	Open	2	Open	2.0	5.5	V	21, 22
Q21		13		12					
Q22		13		12					
Q23		19		15					
Q24		19		15					
Q25		23		27					
Q26		23		27					
Q27		28		32					
Q28		28		32					
Q29		36		40					
Q30		36		40					
Q31		6		2					
Q32		6		2					
Q33		13		12					
Q34		13		12					
Q35		19		15					
Q36		19		15					
Q37		23		27					
Q38	Input transient immunity	23	Open	27	Open	2.0	5.5	V	21, 22

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TABLE XV. Input transient immunity test - Continued.

Test cond	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q39	Input transient immunity	28	Open	32	Open	2.0	5.5	V	21, 22
Q40		28		32					
Q41		36		40					
Q42		36		40					
Q43		6		2					
Q44		6		2					
Q45		13		12					
Q46		13		12					
Q47		19		15					
Q48		19		15					
Q49		23		27					
Q50		23		27					
Q51		28		32					
Q52		28		32					
Q53		36		40					
Q54	Input transient immunity	36	Open	40	Open	2.0	5.5	V	21, 22

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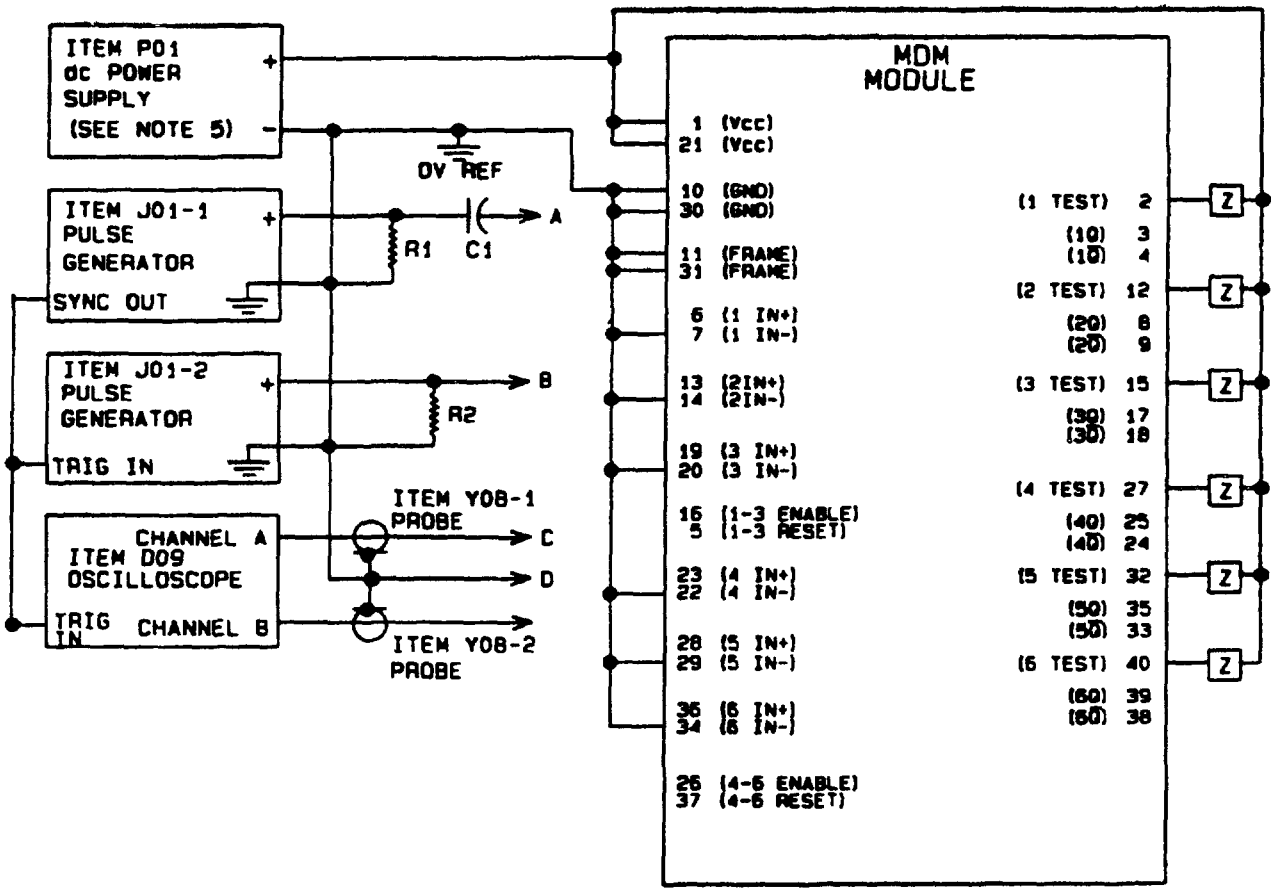


FIGURE 21. Input transient immunity test circuit for table XV.

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NOTES:

1. R1, R2 = 50 Ohms ± 5 percent, 1 W.

2. C1 = 10 μF ± 20 percent, 15 Vdc.

3. Resistor R1 and capacitor C1 are located at the module end of the generator signal cable.

4. Load outputs with load as specified on figure 13.

5. For test conditions Q1-Q19, Q21, Q23, Q25, Q27, Q29, Q31, Q33, Q35, Q37, Q39, Q41, Q43, Q45, Q47, Q49, Q51 and Q53, set item P01 to 4.5 V ± 1 percent. For test conditions Q20, Q22, Q24, Q26, Q28, Q30, Q32, Q34, Q36, Q38, Q40, Q42, Q44, Q46, Q48, Q50, Q52, and Q54, set item P01 to 5.5 V ± 1 percent.

FIGURE 21. Input transient immunity test circuit for table XV - Continued.

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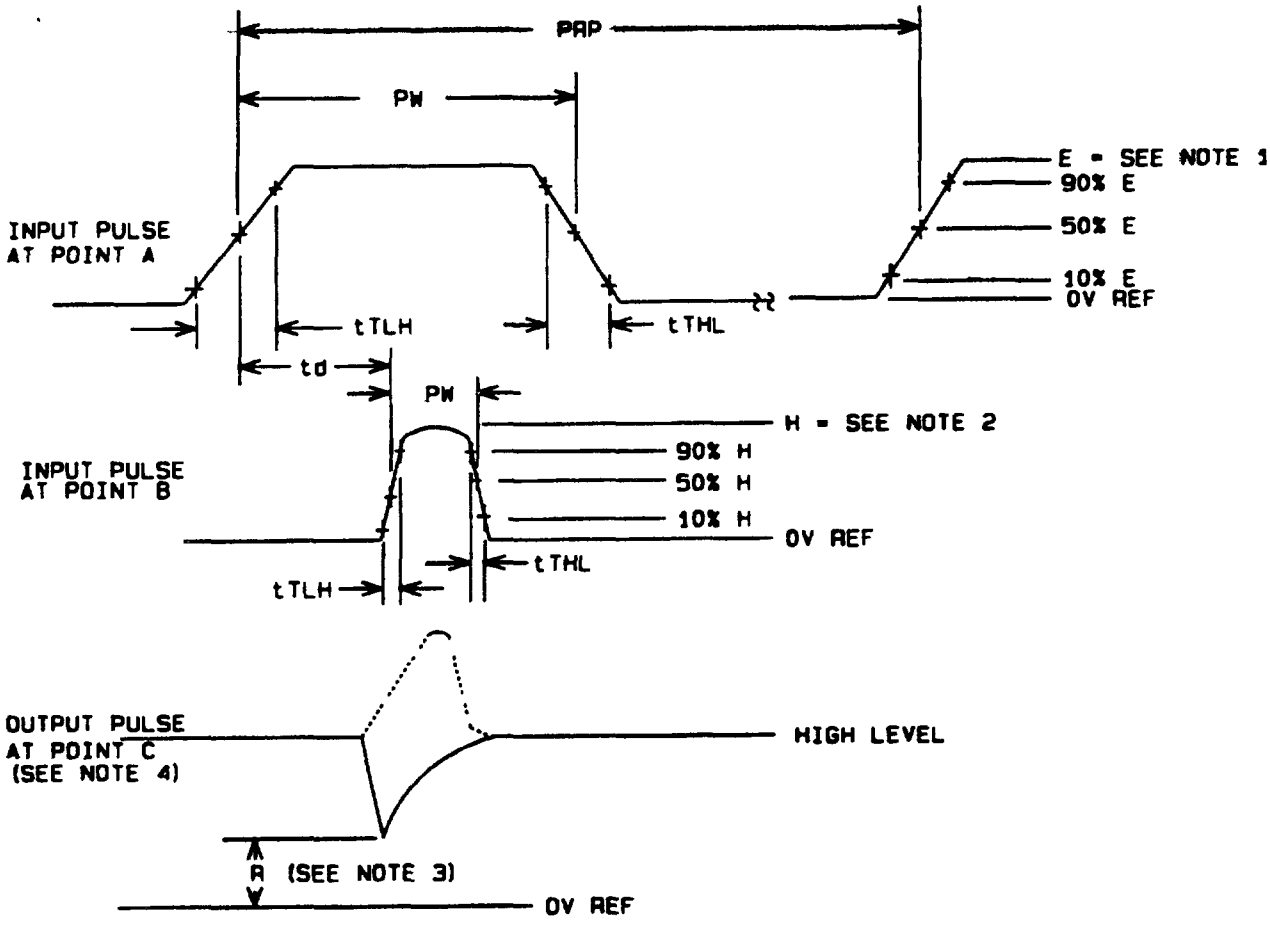


FIGURE 22. Input transient immunity test circuit waveforms for table XV.

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NOTES:

1. Input pulse parameters for item J01-1:

tTLH = tTHL = 20 ± 4 nS.
PRP = 15 ± 1 µS.
PW = For test conditions Q13-Q18, PW = 1.5 µS ± 10 percent.
For test conditions Q19-Q30, PW = 0.25 µS ± 10 percent.
For test conditions Q31-Q42, PW = 0.5 µS ± 10 percent.
For test conditions Q43-Q54, PW = 1.0 µS ± 10 percent.
E = For test conditions Q13-Q18, E = 4.0 V ± 10 percent.
For test conditions Q19-Q30, E = 2.0 V ± 10 percent.
For test conditions Q31-Q42, E = 1.0 V ± 10 percent.
For test conditions Q43-Q54, E = 0.5 V ± 10 percent.

2. Input pulse parameters for item J01-2:

tTLH = tTHL = 6 ± 1 nS.
td = 1.5 ± 0.2 µS.
PW = 20 ± 2 nS.

H =

H ± 50 mV (V)	Temp °C
1.20	0
1.10	25
.95	60
.80	80

3. Value of R must conform to the requirements of test condition Q.

4. If circuit triggering does not occur (negative spike), some circuit feed-through of transient (input pulse at point B) may be observed (dashed waveform).

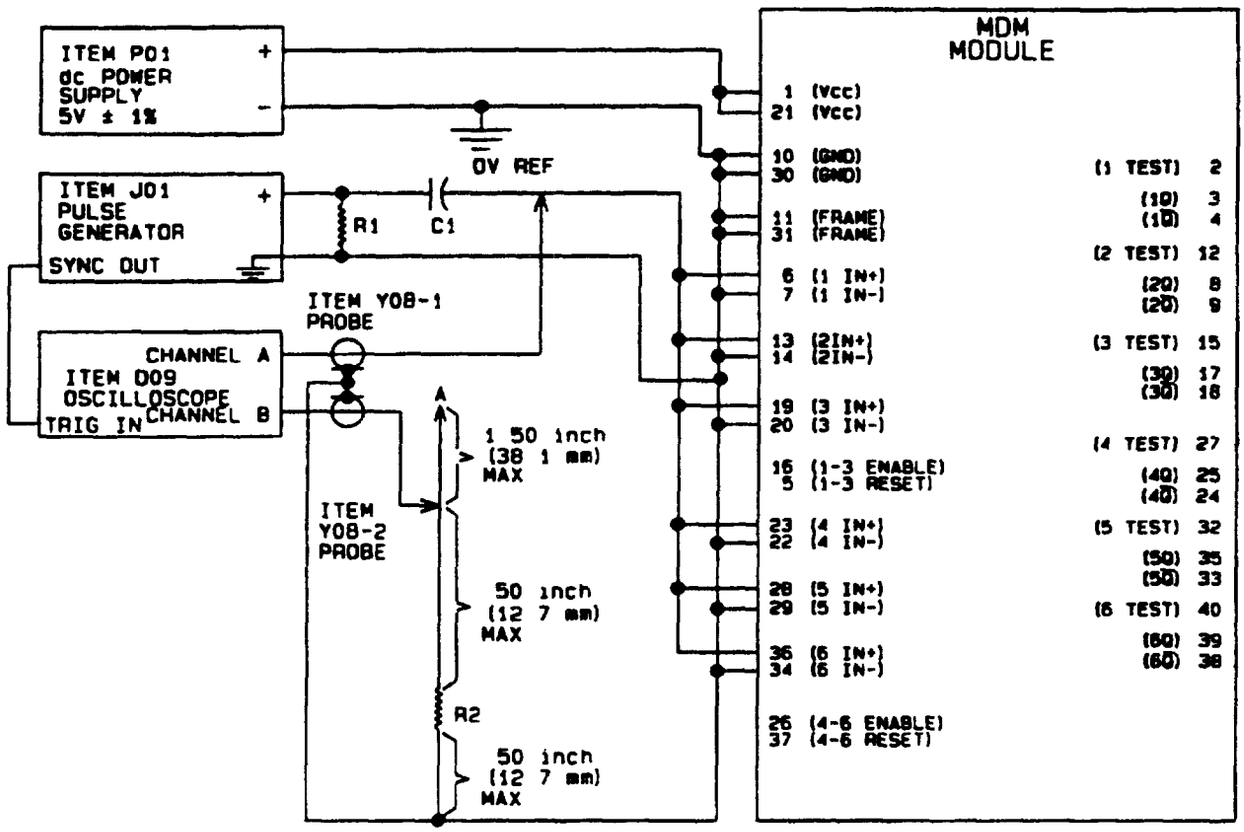
FIGURE 22. Input transient immunity test circuit waveforms for table XV - Continued.

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TABLE XVI. Input crosstalk test.

Test cond	Parameter	Connect	Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	Min	Max		
R1	Input crosstalk	16	0.0	0.6	Vp-p	23, 24
R2		26				
R3		5				
R4	Input crosstalk	37	0.0	0.6	Vp-p	23, 24

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
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NOTES:

1. C1 = 10 μF ± 20 percent, 15 Vdc.

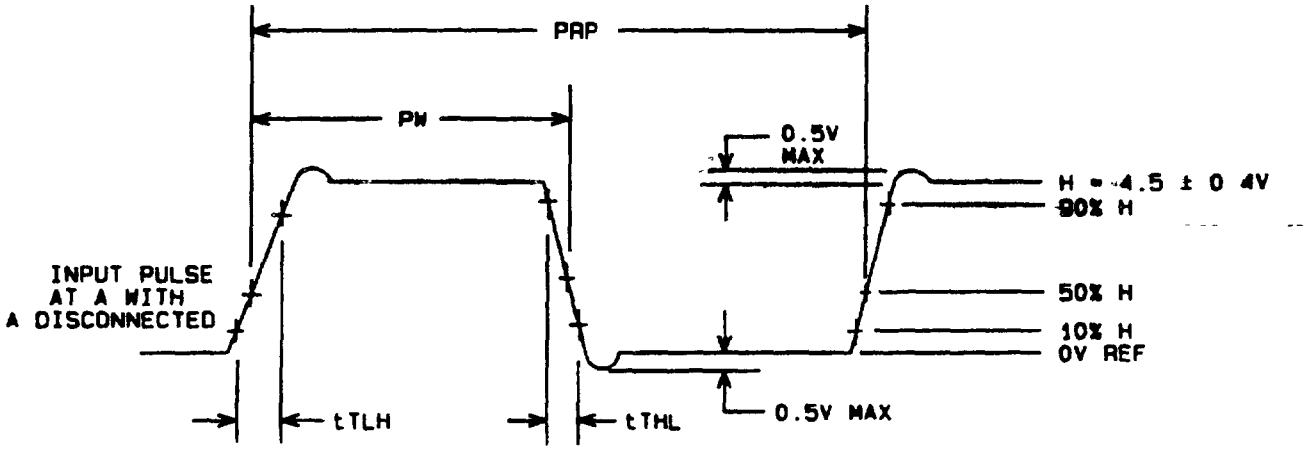
2. R1 = 50 Ohms ± 5 percent, 1 W.

3. R2 = 154 Ohms ± 1 percent, 1/2 W.

4. Resistor R1 and capacitor C1 are located at the module end of the generator signal cable.

FIGURE 23. Input crosstalk test circuit for table XVI.

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NOTES:

1. Input pulse parameters:

   PW = 500 ± 100 nS.
   tTLH = tTHL = 10 ± 2 nS.
   PRP = 6 ± 0.5 µS.

2. Item Y08-2 and its associated ground lead must be within 1/2 inch (12.7 mm) from the body of the resistor R2.

3. At each test point, the peak-to-peak voltage developed across resistor R2, condition R must conform to the requirements of test condition R.

4. The length of the conductor from the body of resistor R2 to point A shall not exceed 2 inches (50.8 mm).

5. The length of the conductor from the body of resistor R2 to the junction of pins 10, 11, 31, and 30 shall not exceed 2 inches (50.8 mm).

6. The maximum capacitance between adjacent input connectors on the test fixture shall not exceed 10 pF when measured at 1 MHz ± 10 percent with the module removed from the test fixture.

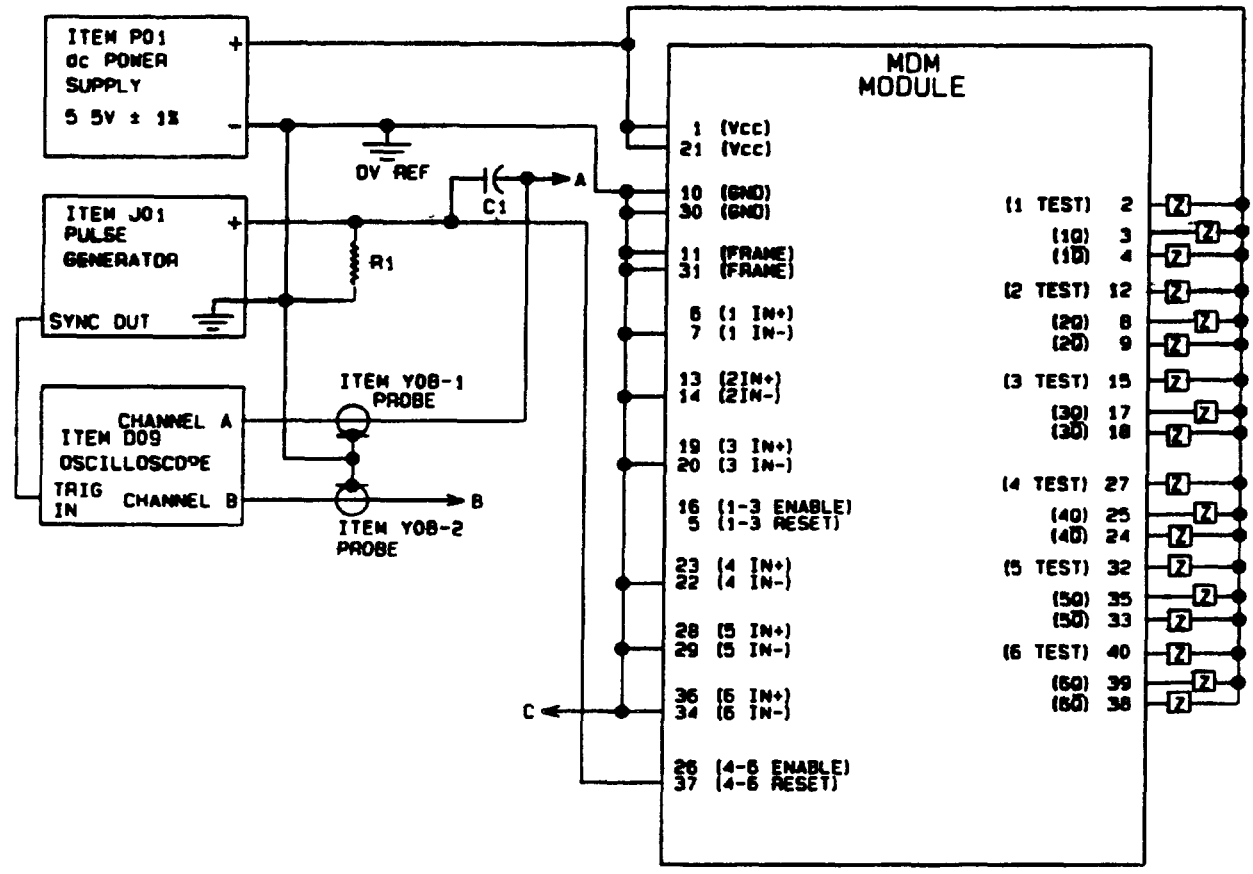
FIGURE 24. Input crosstalk test circuit waveform for table XVI.

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TABLE XVII. Output crosstalk test.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Min	Max		
S1	Output crosstalk	13,19,23,28, 36	2	Open	0.0	0.8	V	25, 26
S2			4	Open				
S3		13,19,23,28, 36	3	2				
S4		6,19,23,28,36	12	Open				
S5			9	Open				
S6		6,19,23,28,36	8	12				
S7		6,13,23,28,36	15	Open				
S8			18	Open				
S9		6,13,23,28,36	17	15				
S10		6,13,19,28,36	27	Open				
S11			24	Open				
S12		6,13,19,28,36	25	27				
S13		6,13,19,23,36	32	Open				
S14			33	Open				
S15		6,13,19,23,36	35	32				
S16		6,13,19,23,28	40	Open				
S17			38	Open				
S18	Output crosstalk	6,13,19,23,28	39	40	0.0	0.8	V	25, 26

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
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NOTES:

1. R1 = 50 Ohms ± 5 percent, 1 W.

2. C1 = 10 μF ± 20 percent, 15 Vdc.

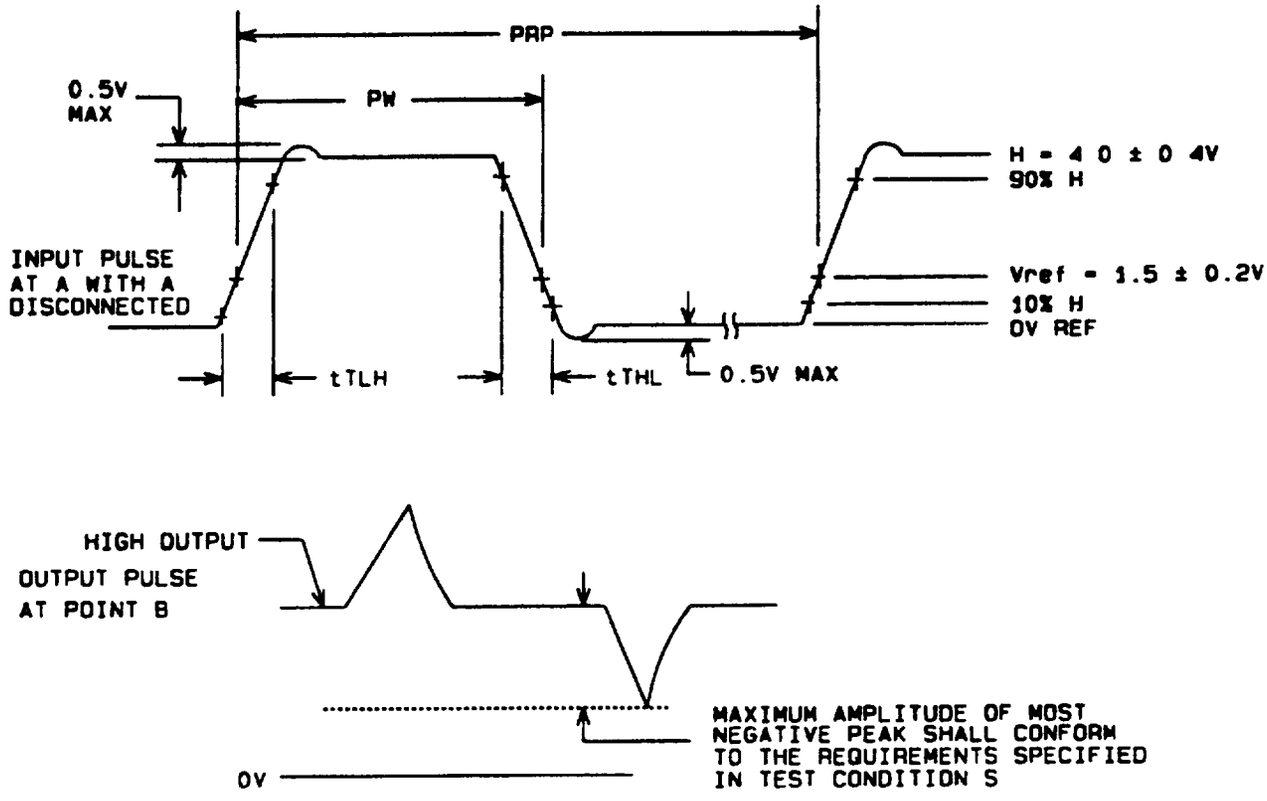
3. Load all outputs with load as specified on figure 13.

4. Pins connected to point B shall have the load removed before making the measurement. The load should be reconnected before proceeding to the next test condition.

5. Resistor R1 and capacitor C1 are located at the module end of the generator signal cable.

FIGURE 25. Output crosstalk test circuit for table XVII.

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NOTE:

1. **Input pulse parameters:**

   PW = 1.0 ± 0.1 µS.
   PRP = 6.0 ± 0.5 µS.
   tTLH = tTHL = 20 ± 5 nS.

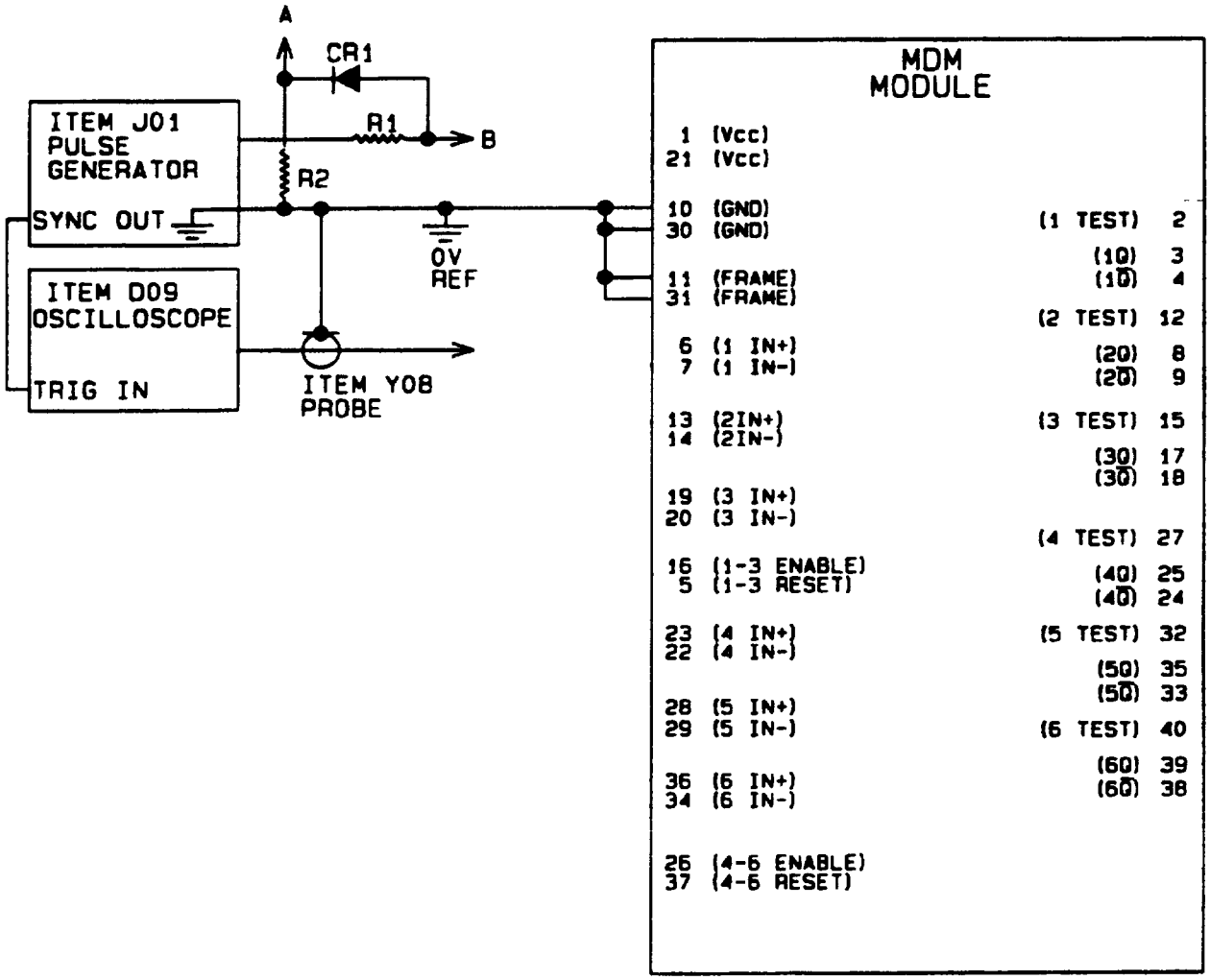
FIGURE 26. **Output crosstalk test circuit waveforms for table XVII.**

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TABLE XVIII. Power supply transient rise time test.

Test cond	Parameter	Connect B to pin	Initial limits 25° ± 5°C Ambient		Unit	Test fig.
			Min	Max		
T1	Power supply transient rise time	1	20	136	µS	27, 28
T2	Power supply transient rise time	21	20	136	µS	27, 28

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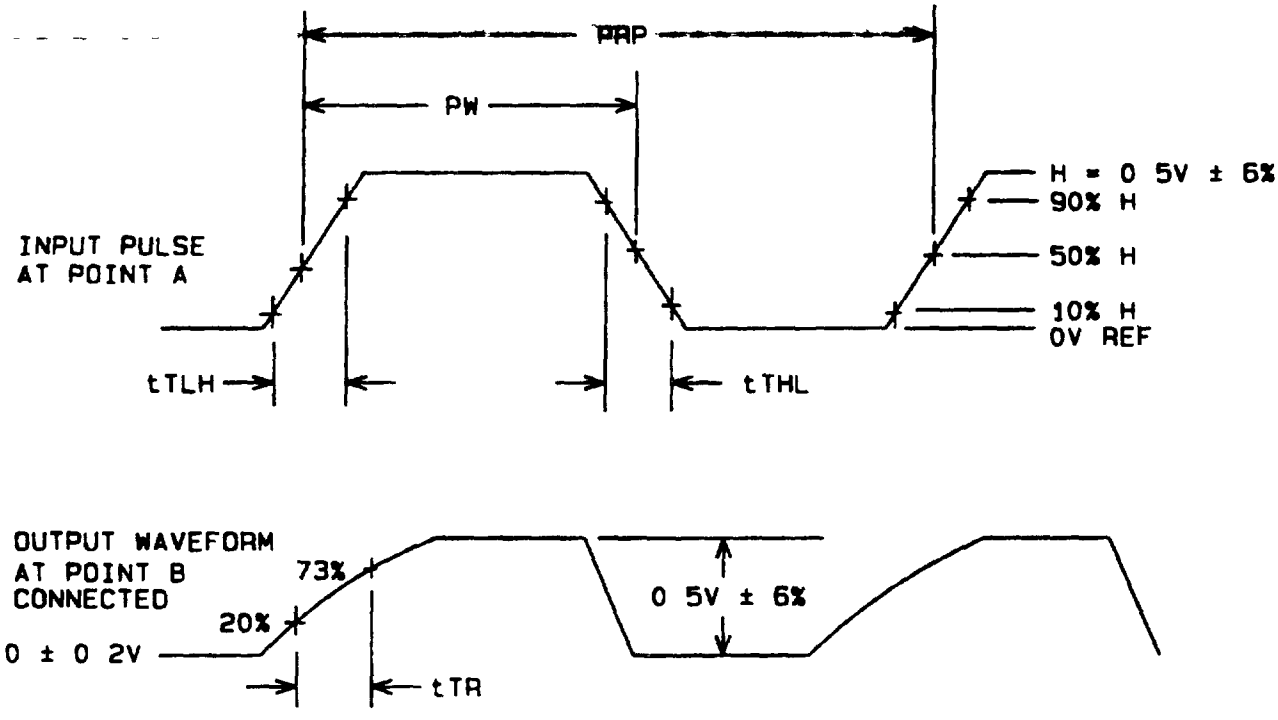


NOTES:

1. CR1 = 1N4150 or equivalent.
2. R1 = 464 Ohms ± 1 percent, 1/4 W.
3. R2 = 50 Ohms ± 5 percent, 1 W.

FIGURE 27. Power supply transient rise time test circuit for table XVIII.

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NOTE:

1. Input pulse parameters:

   PW = 750 µS ± 5 percent.
   PRP = 1.5 mS ± 5 percent.
   tTLH = tTHL = 50 nS ± 10 percent.

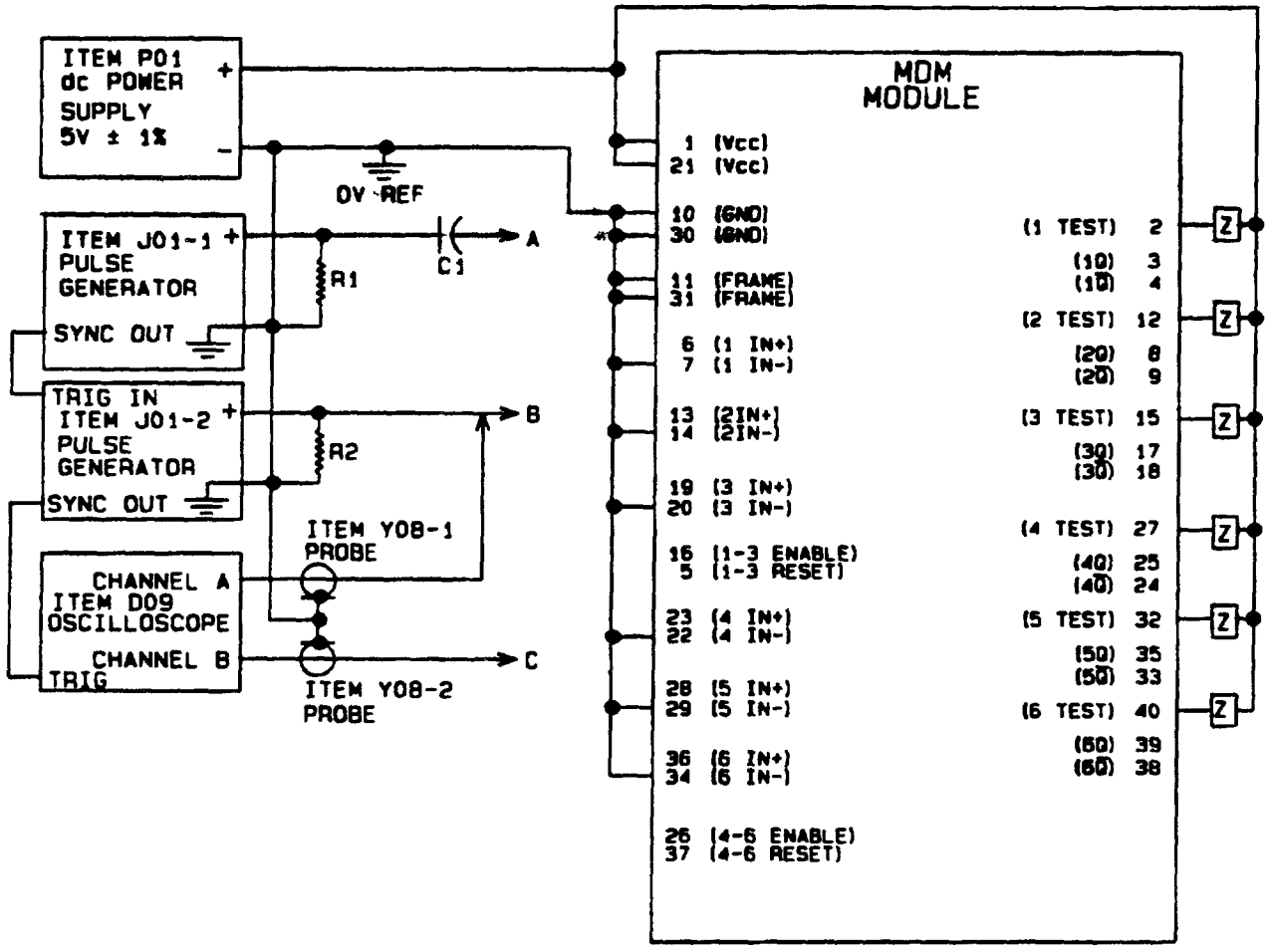
FIGURE 28. Power supply transient rise time test circuit waveforms for table XVIII.

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TABLE XIX. Enable time delay (tdON and tdOFF) test.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Min	Max		
U1	1-3 Enable time delay tdON	6	16	2	0	870	nS	29,30
U2		13		12				
U3	1-3 Enable time delay tdON	19	16	15				
U4	4-6 Enable time delay tdON	23	26	27				
U5		28		32				
U6	4-6 Enable time delay tdON	36	26	40		870		
U7	1-3 Enable time delay tdOFF	6	16	2		100		
U8		13		12				
U9	1-3 Enable time delay tdOFF	19	16	15				
U10	4-6 Enable time delay tdOFF	23	26	27				
U11		28		32				
U12	4-6 Enable time delay tdOFF	36	26	40	0	100	nS	29,30

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NOTES:

1. R1, R2 = 50 Ohms ± 5 percent, 1 W.

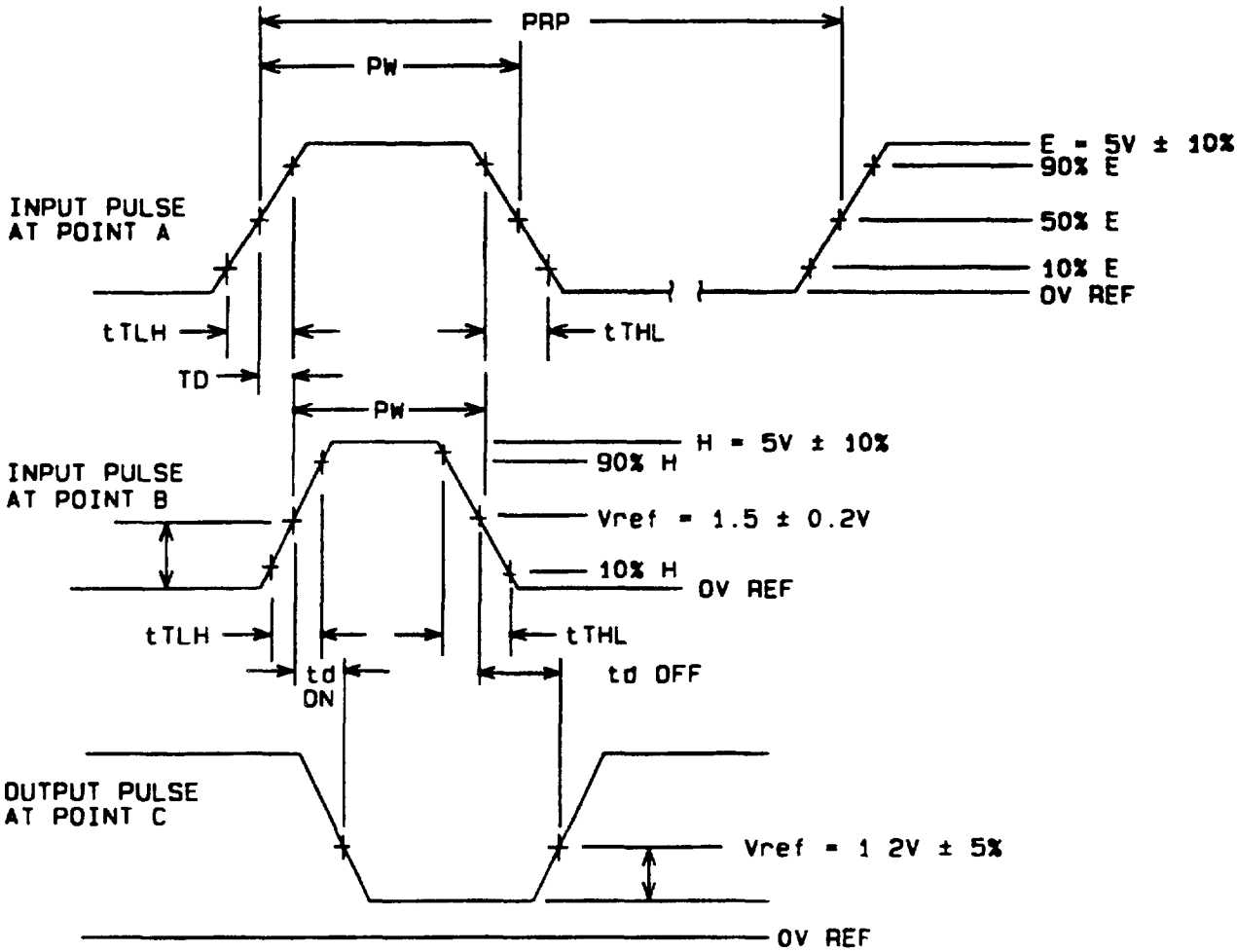
2. C1 = 10 μF ± 20 percent, 15 Vdc.

3. Resistor R1 and capacitor C1 are located at the module end of the generator signal cable.

4. Load outputs with load as specified on figure 13.

FIGURE 29. Enable time delay (tdON and tdOFF) test circuit for table XIX.

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NOTES:

1. Input pulse parameters at point A:

   tTLH = tTHL = 40 ± 10 nS.
   PW = 2.0 ± 0.1 µS.
   PRP = 20 µS ± 10 percent.

2. Input pulse parameters at point B:

   tTLH = tTHL = 20 ± 4 nS.
   PW = 1.5 ± 0.1 µS.
   td = 0.2 ± 0.1 µS.

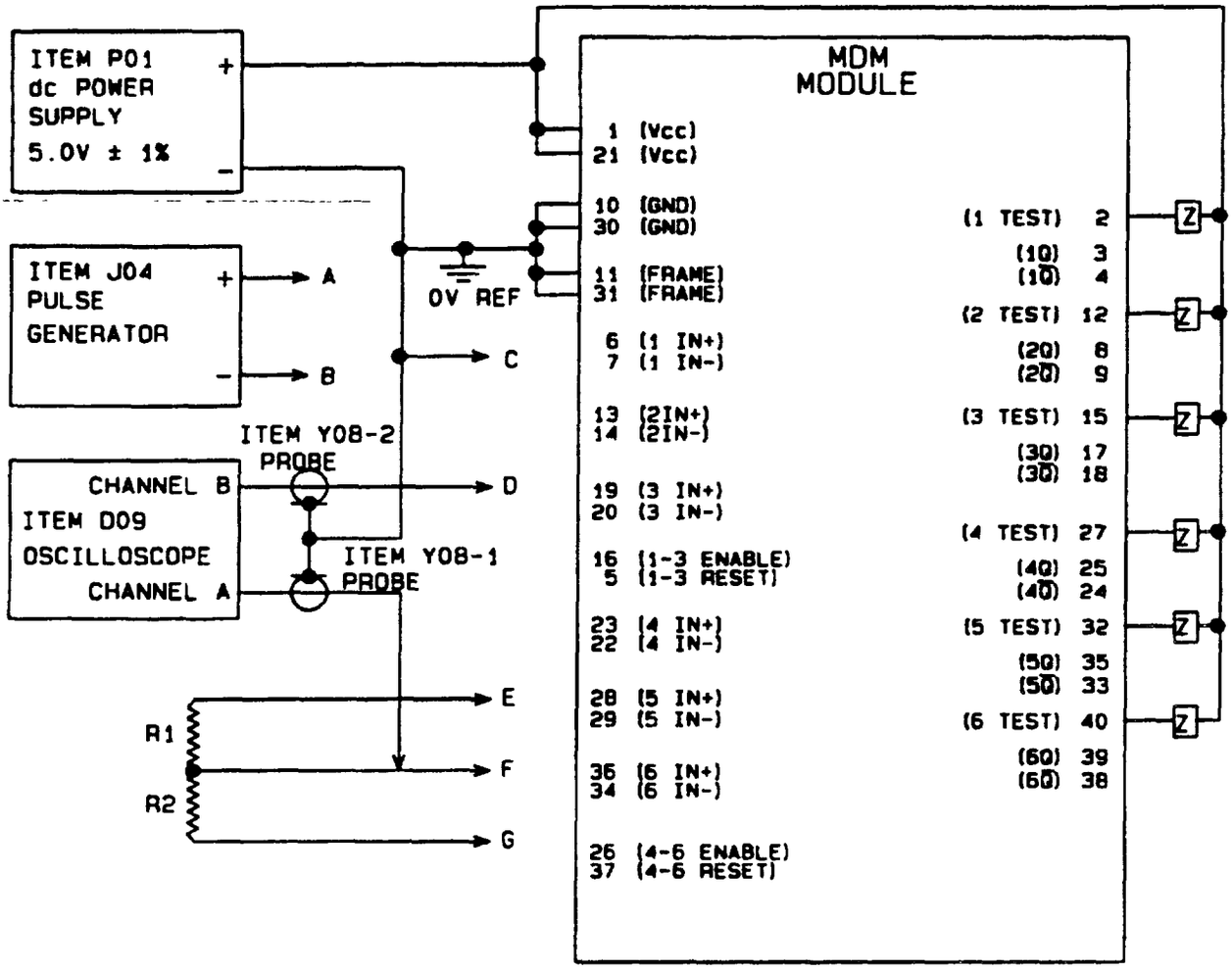
FIGURE 30. Enable time delay (tdON and tdOFF) test circuit waveforms for table XIX.

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TABLE XX. Input noise immunity (common mode) test.

Test cond	Parameter	Connect					Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
V1	1IN Input noise immunity (common mode)	B	2	6	A	7	2.0	5.5	V	31, 32
V2	2IN		12	13		14				
V3	3IN		15	19		20				
V4	4IN		27	22		23				
V5	5IN		32	28		29				
V6	6IN	B	40	34	A	36				
V7	1IN	A	2	6	B	7				
V8	2IN		12	13		14				
V9	3IN		15	19		20				
V10	4IN		27	22		23				
V11	5IN		32	28		29				
V12	6IN Input noise immunity (common mode)	A	40	34	B	36	2.0	5.5	V	31, 32

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NOTES:

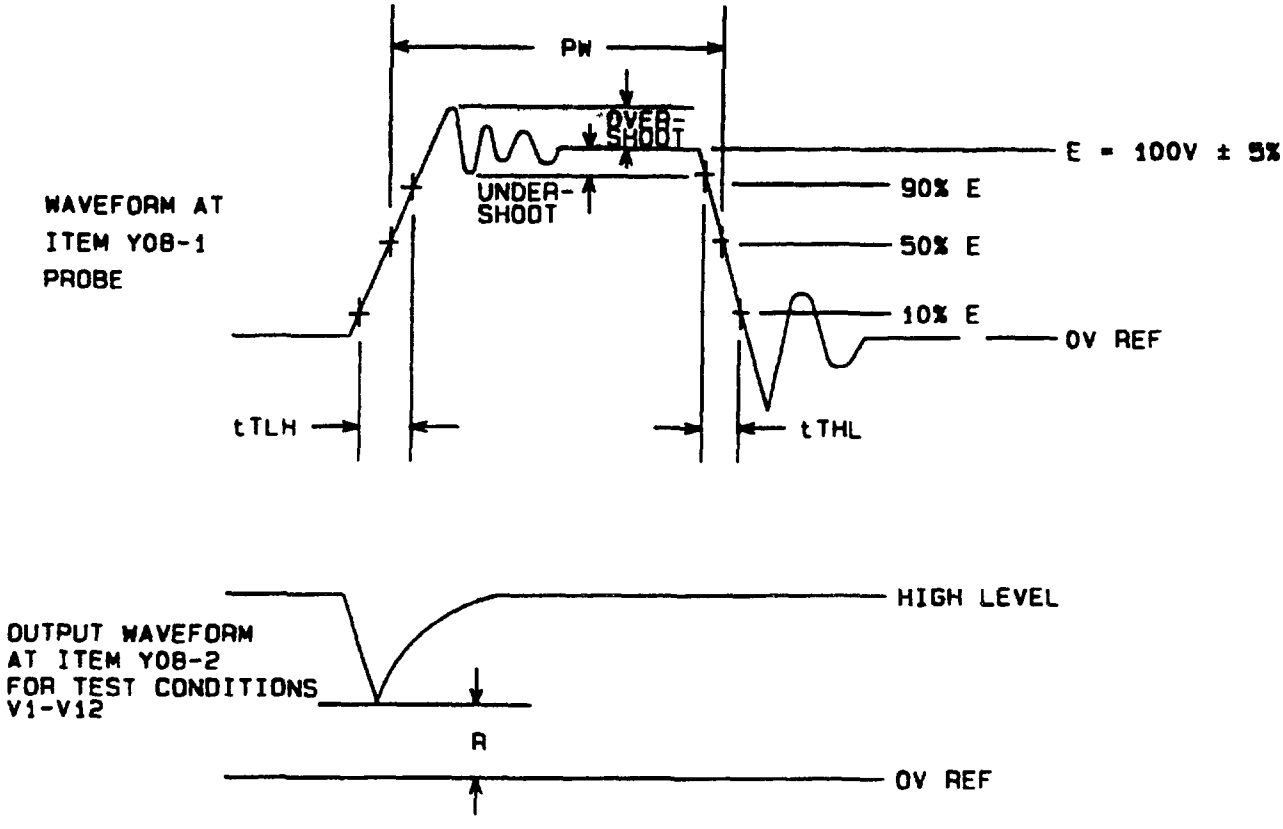
1. R1, R2 = 51.1 Ohms ± 1 percent, 1/4 W.

2. Load outputs with load as specified on figure 13.

3. Maximum loop length from pin 1 to each output, including the length of the body of the resistor, shall not exceed 4.5 inches (114.3 mm).

FIGURE 31. Input noise immunity (common mode) test circuit for table XX.

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NOTE:

1. Input pulse parameters (using item D05):

   tTLH = tTHL = 10 ± 10 nS.
   PW = 10 µS minimum, 10 mS maximum.
   Overshoot = undershoot = 25 V maximum.
   Duty cycle = 10 percent maximum.

FIGURE 32. Input noise immunity (common mode) test circuit waveforms for table XX.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-M-28787/110B	2 DOCUMENT DATE (YYMMDD) 1994 May 25

3 DOCUMENT TITLE
Modules, Standard Electronic, Receiver, Interface, Digital, Key Code MDM

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a NAME Technical Point of Contact (TPOC): Mr. Joseph P. Toy (Code 603A) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b TELEPHONE (Include Area Code) (1) Commercial TPOC: 812-854-5993	(2) AUTOVON 88-482-5993
c ADDRESS (Include Zip Code) Commander ATTN: Joseph P. Toy, Code 603A, Bldg. 2917 NAVSURFWARCENDIV, 300 Highway 361 Crane, IN 47522-5001	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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